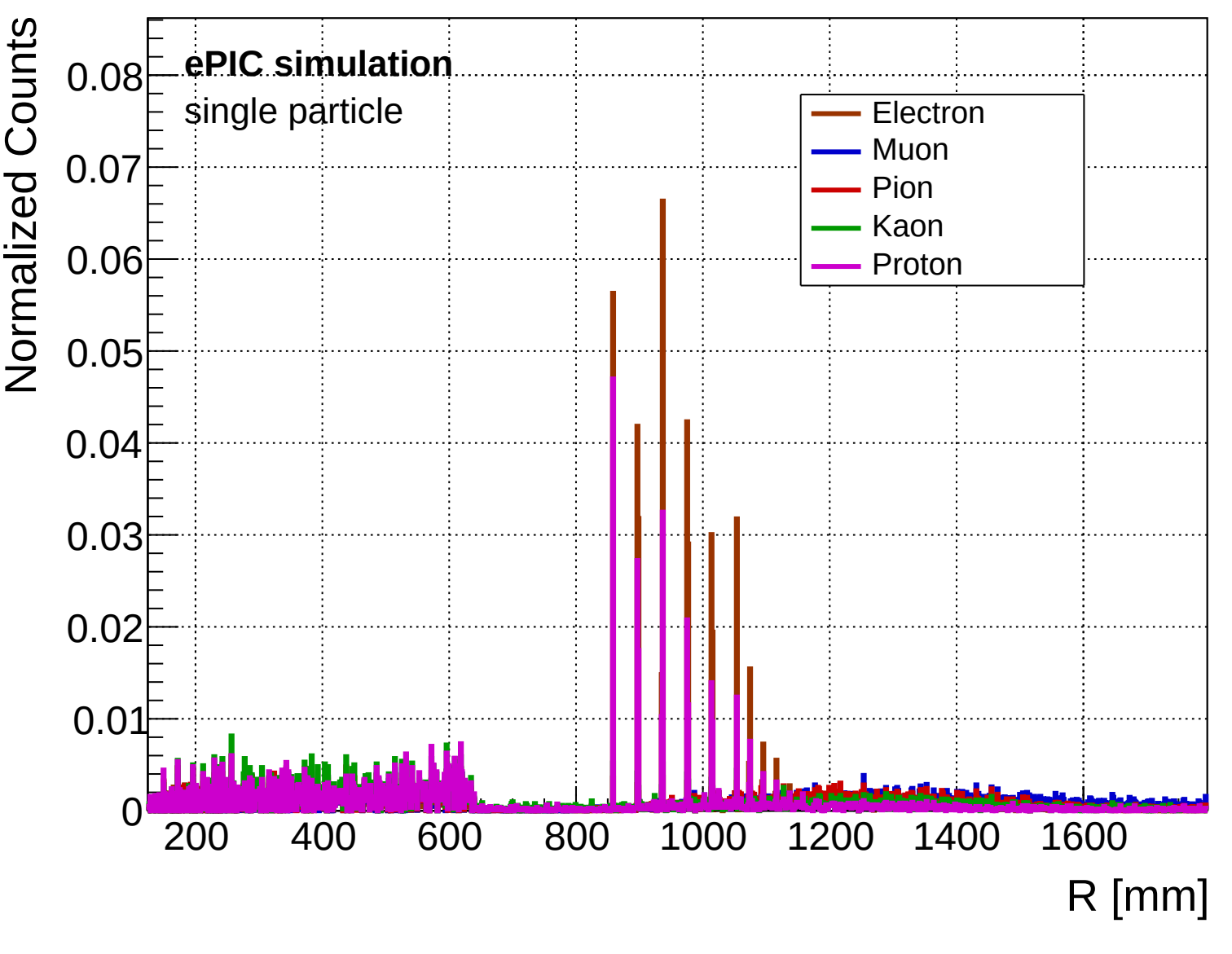
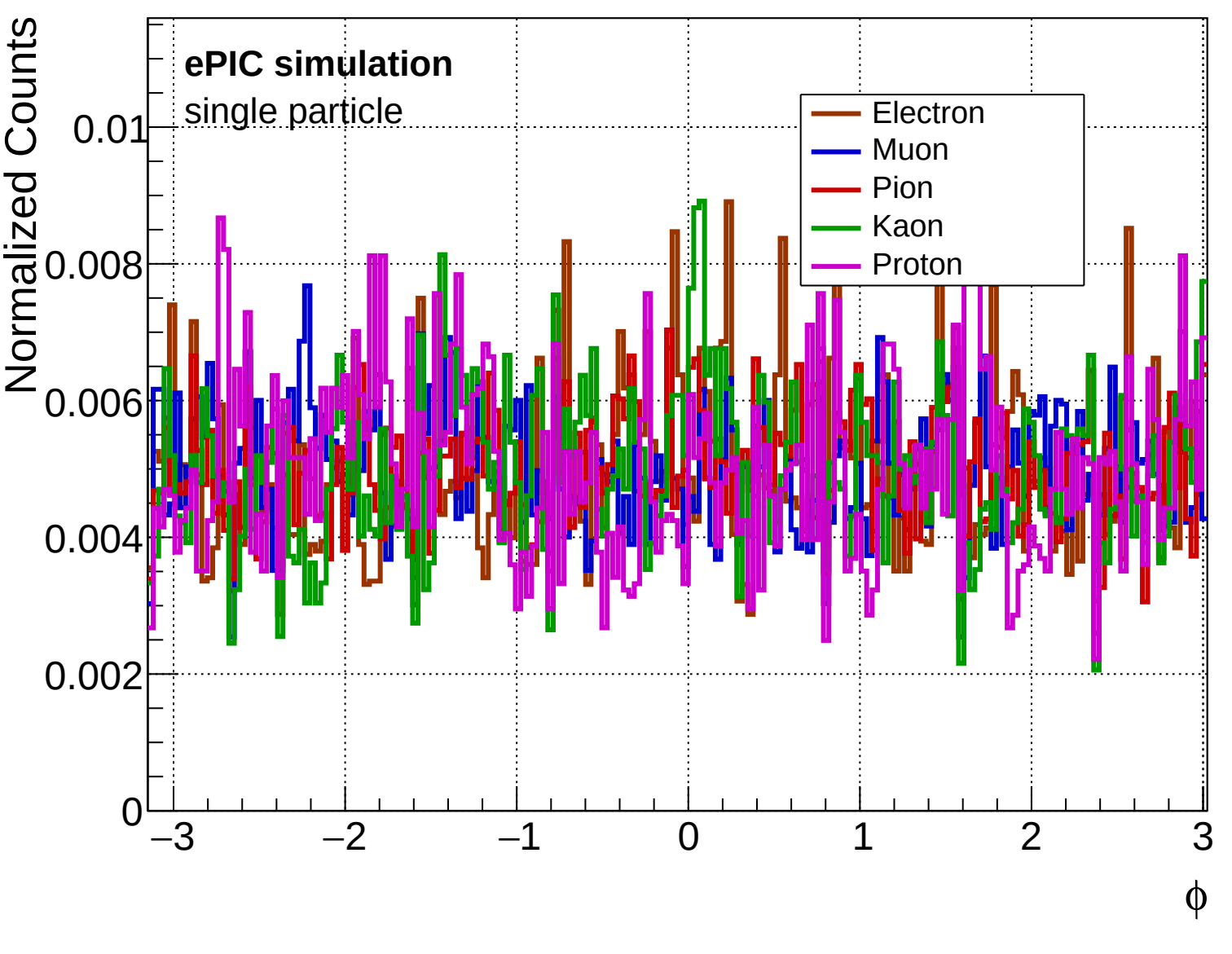
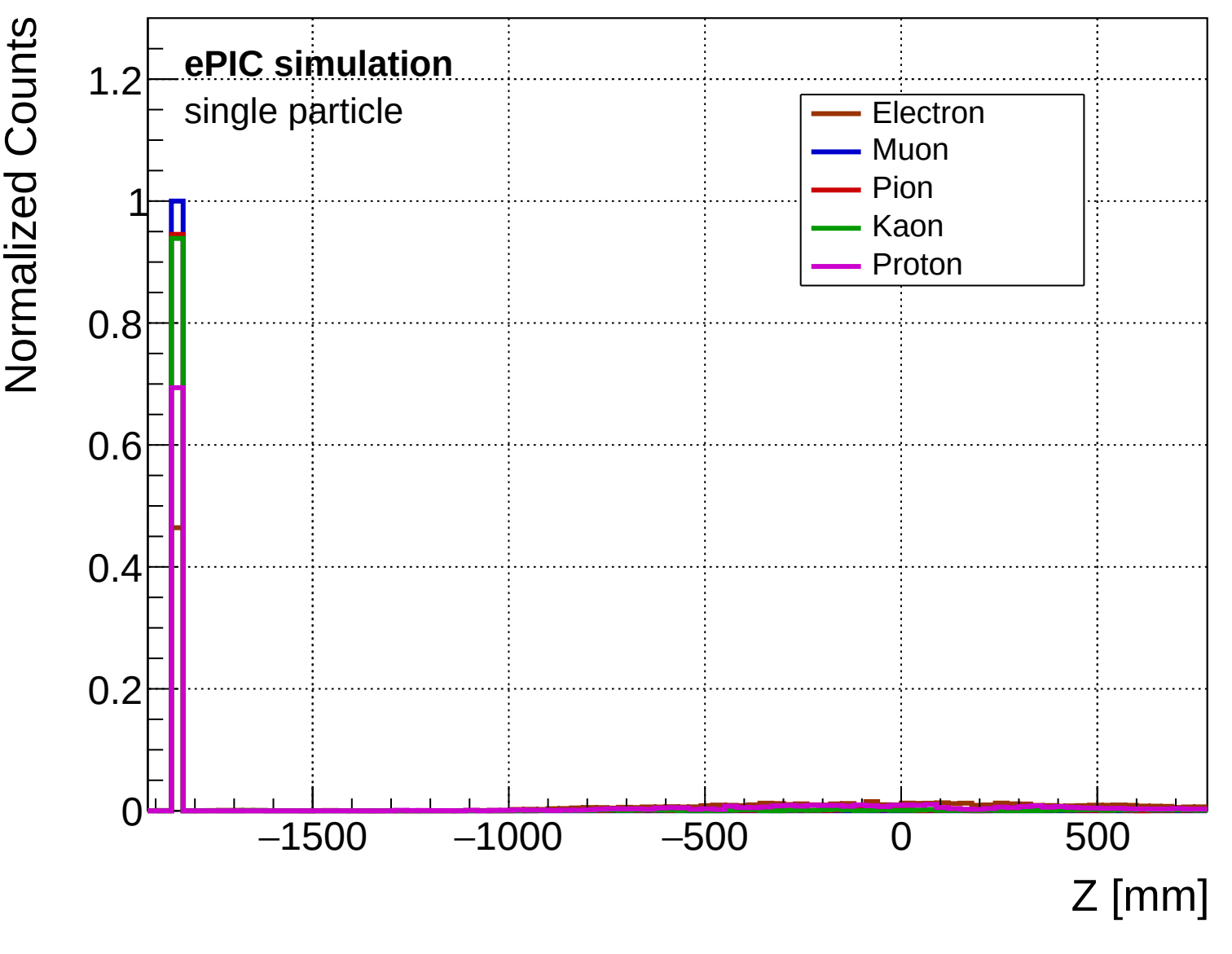
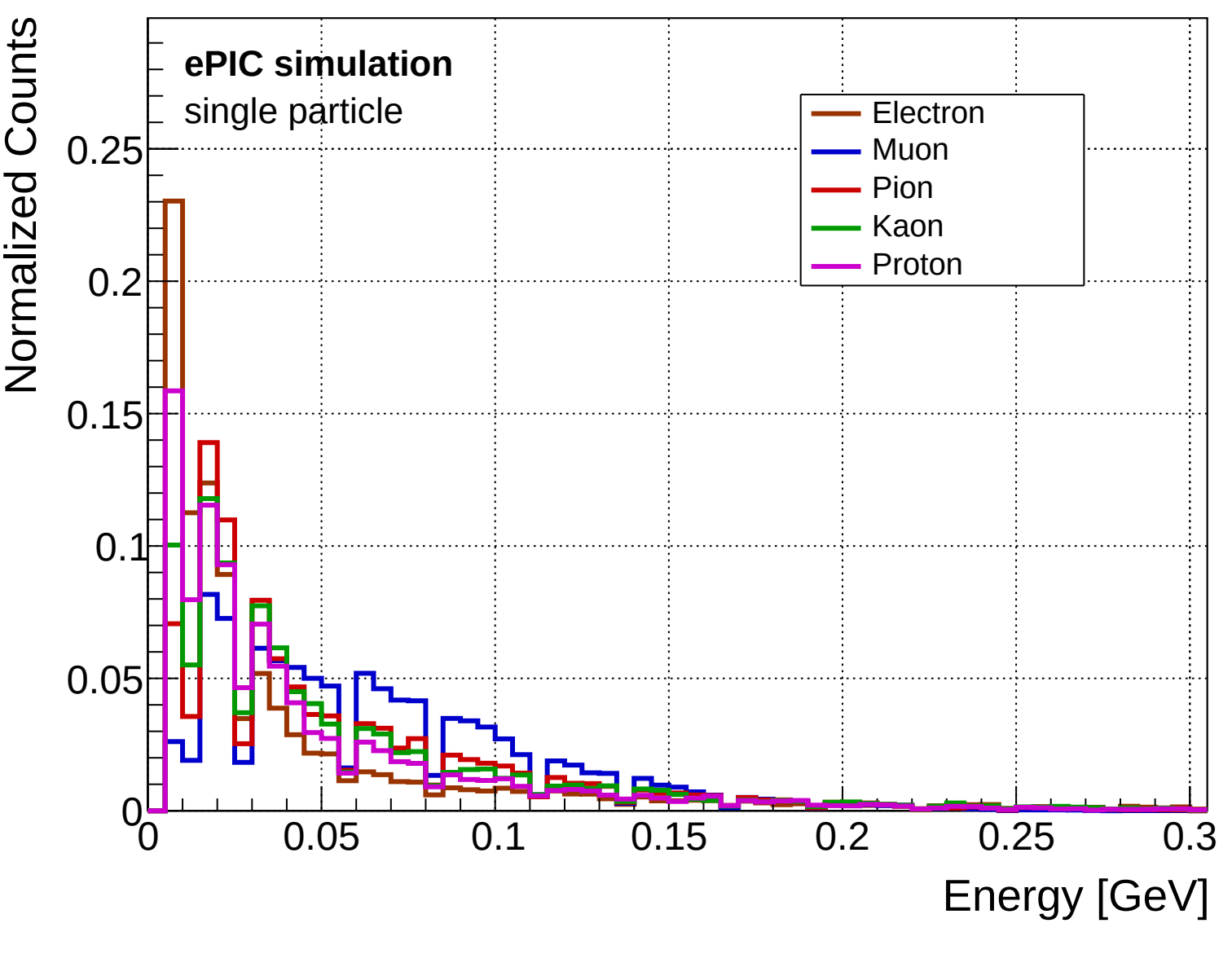
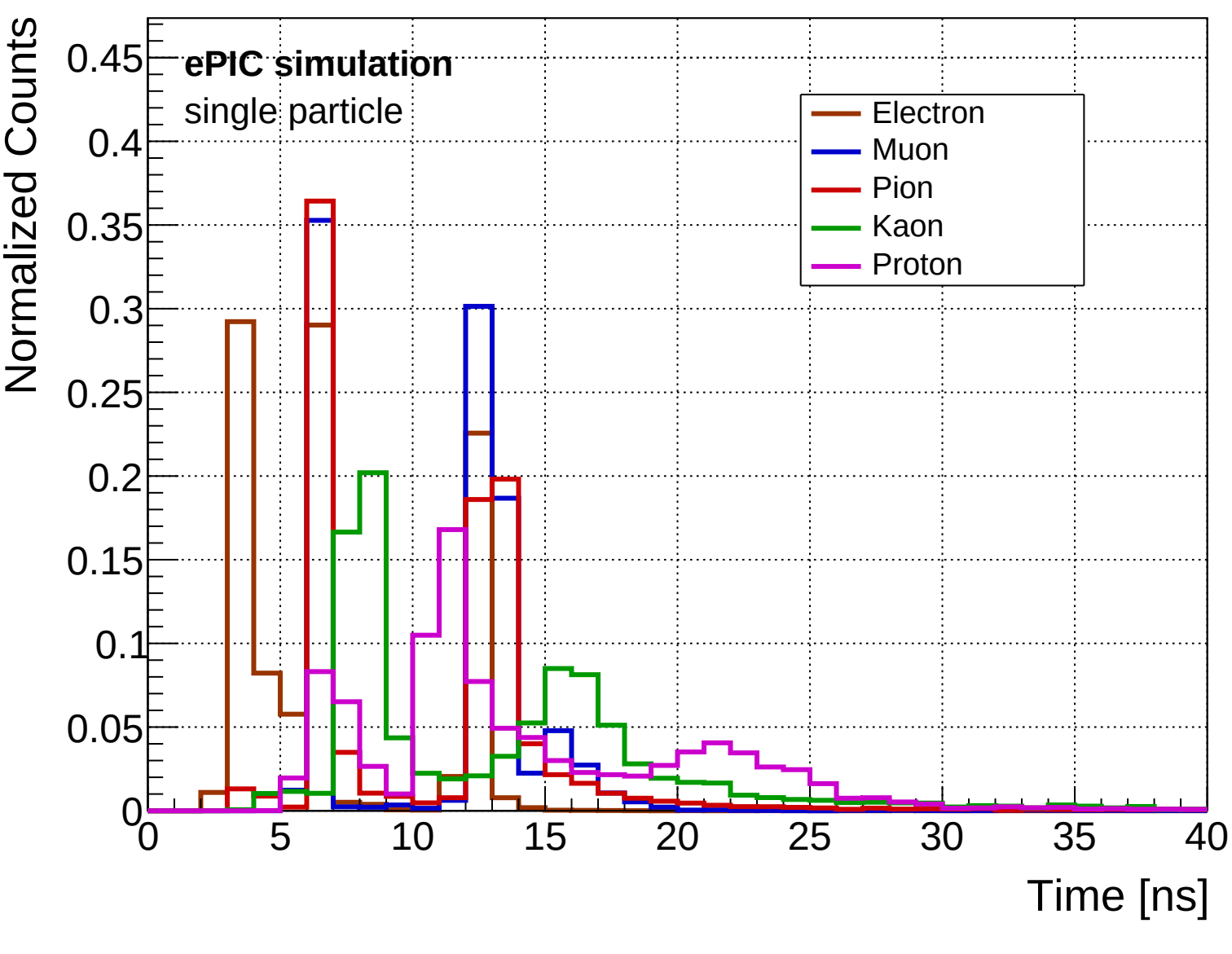
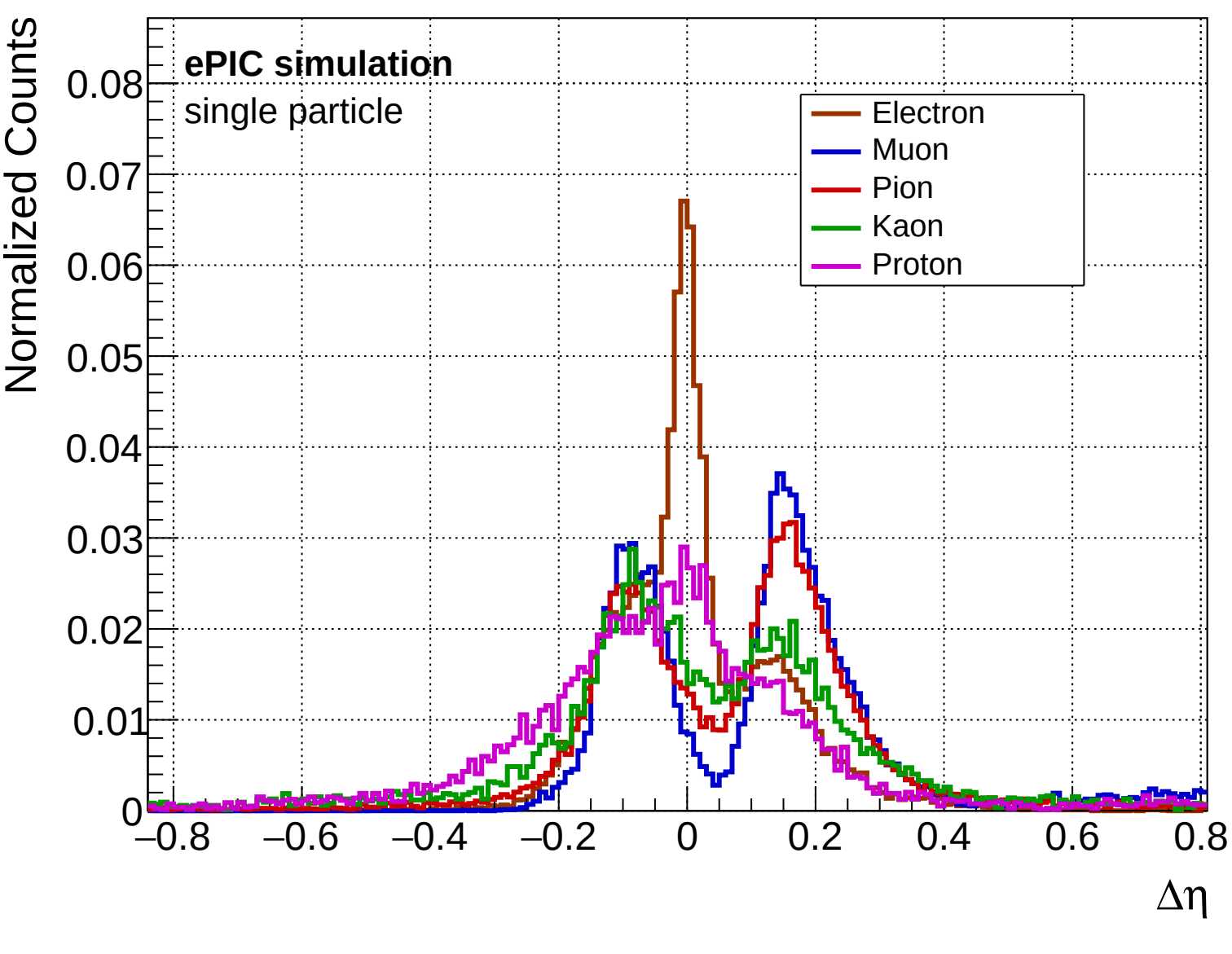

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
200
400
600
800
1000
1200
1400
1600
R [mm]

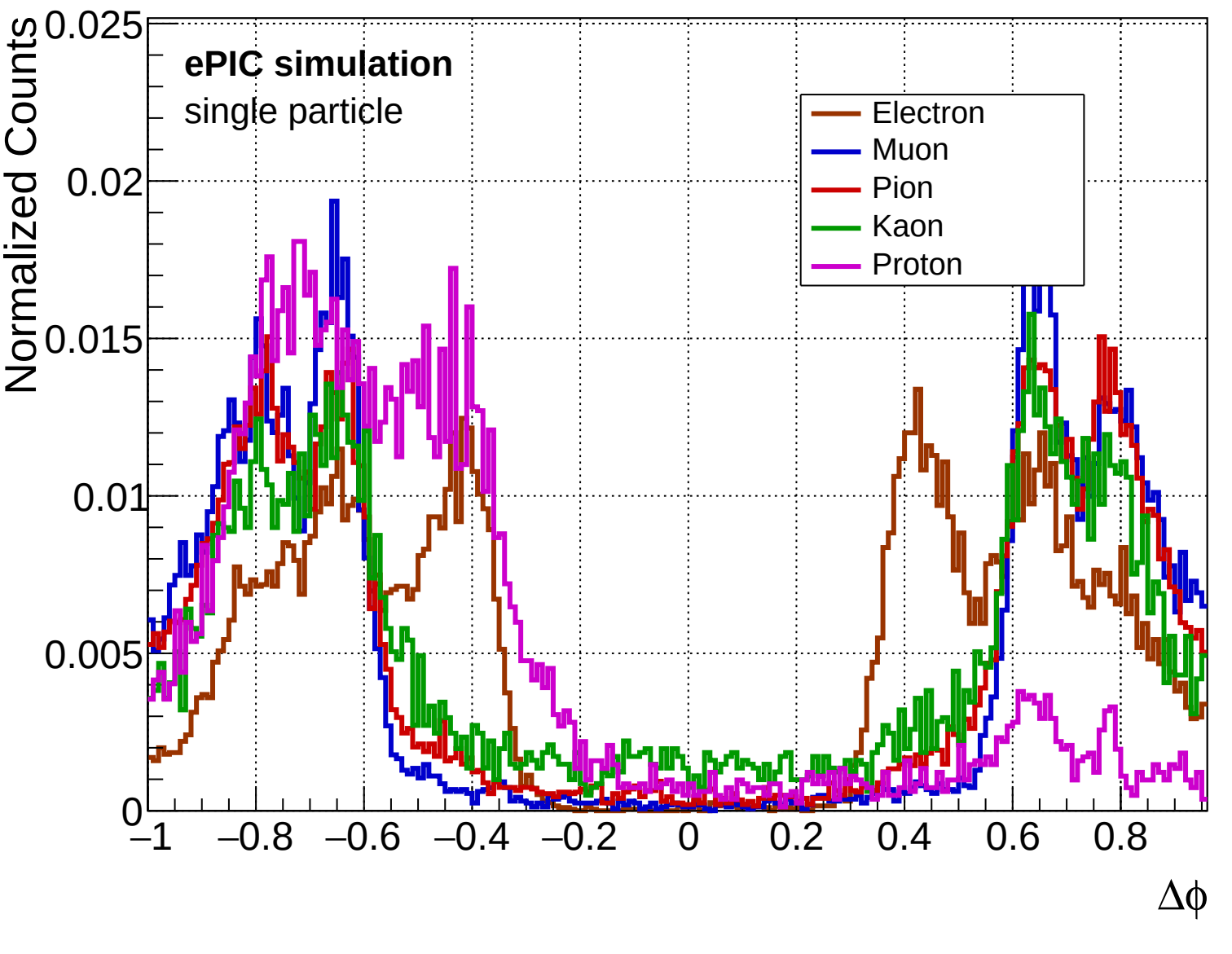

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.01
0.008
0.006
0.004
0.002
0
−3
−2
−1
0
1
2
3
ϕ

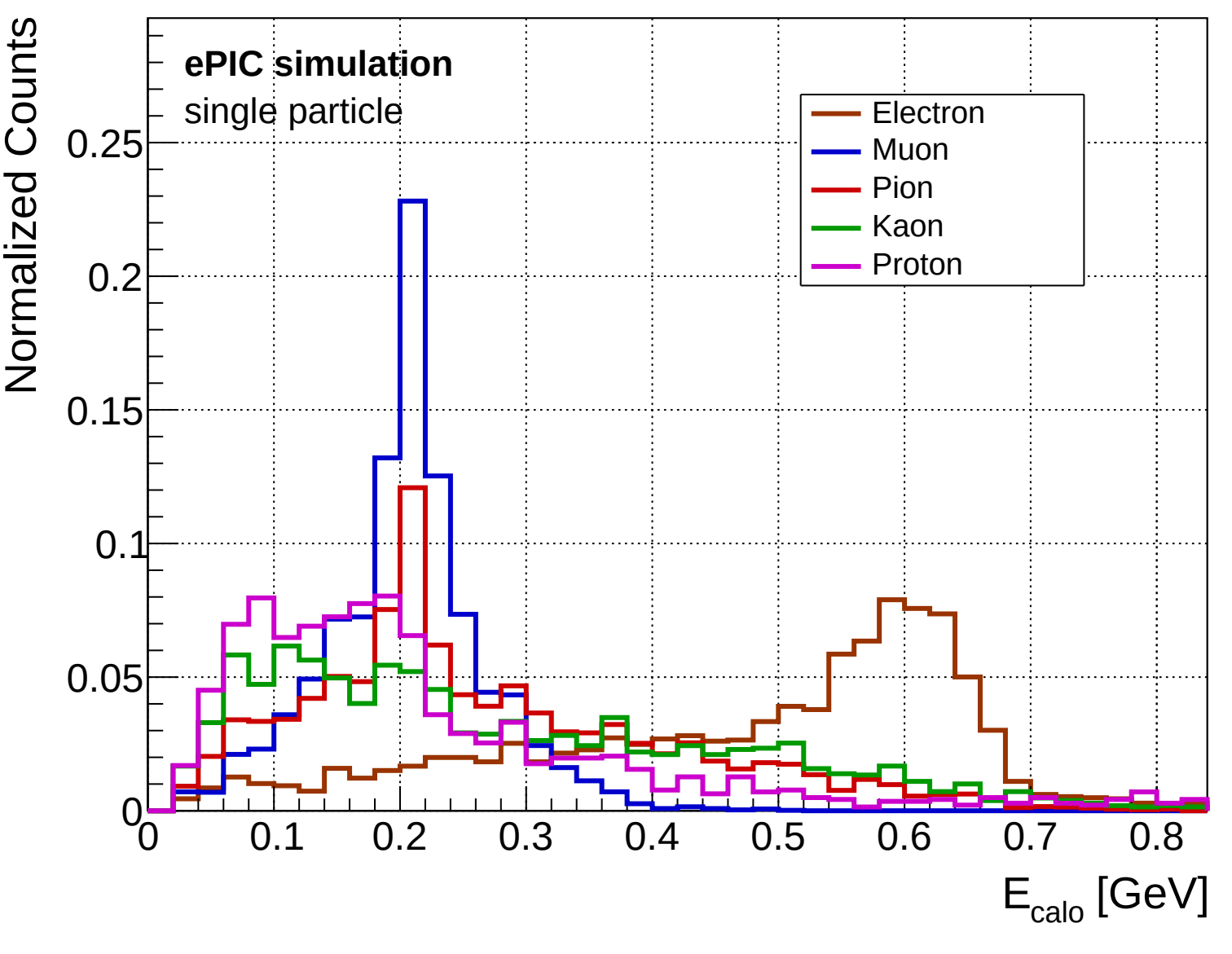

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
1.2
1
0.8
0.6
0.4
0.2
0
–1500
–1000
–500
0
500
Z [mm]

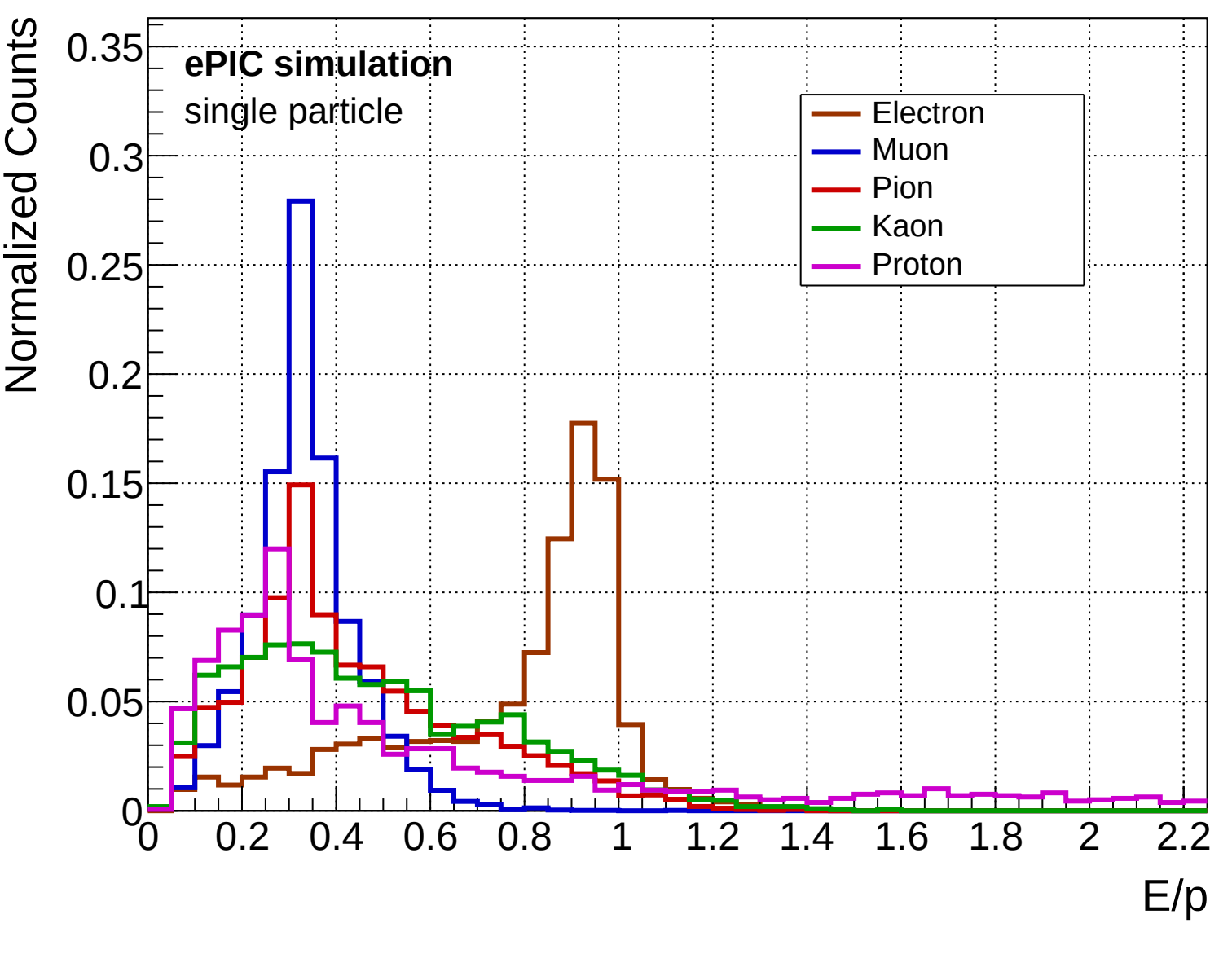

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
Energy [GeV]
0
0.05
0.1
0.15
0.2
0.25
0.3

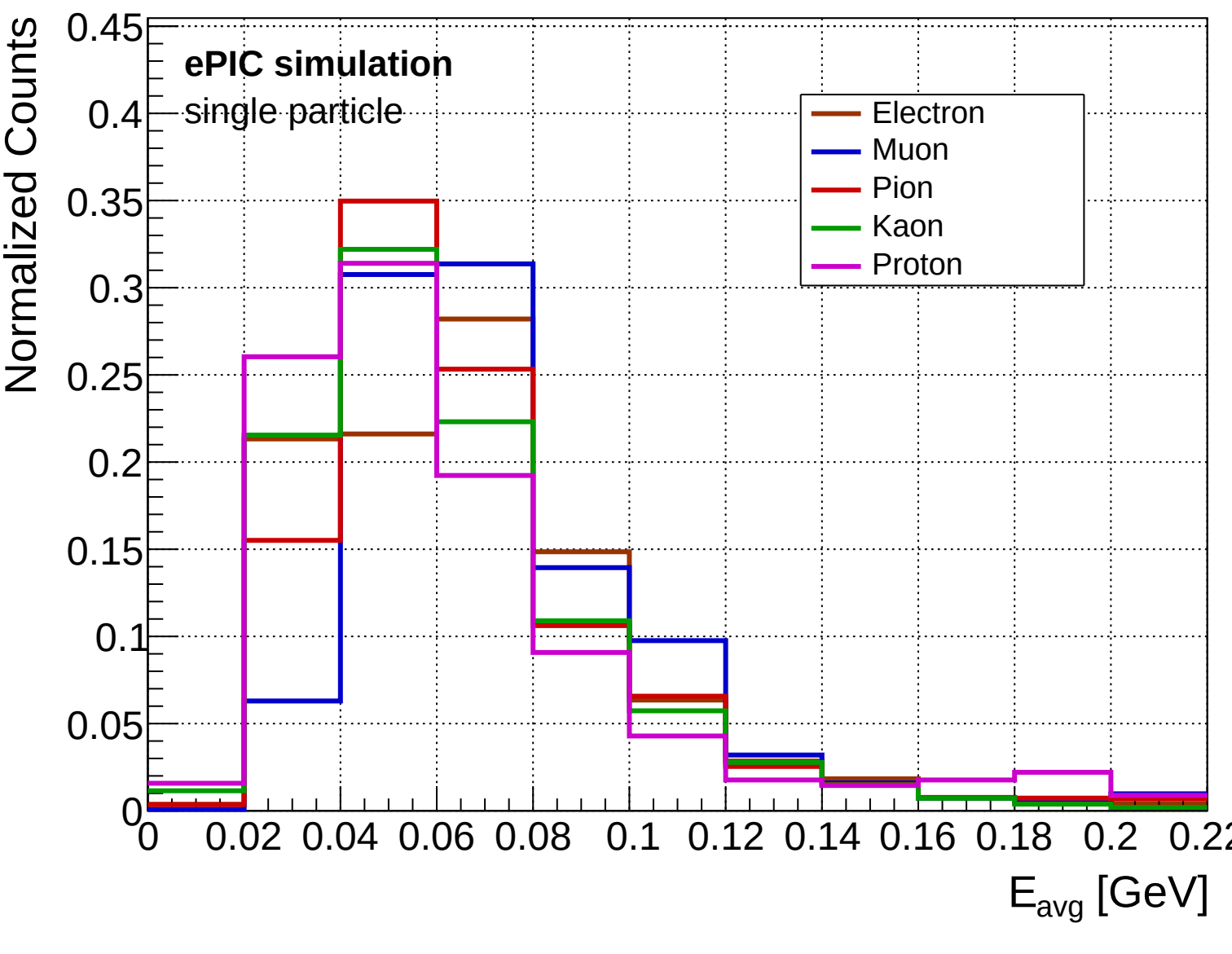

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
5
10
15
20
25
30
35
40
Time [ns]

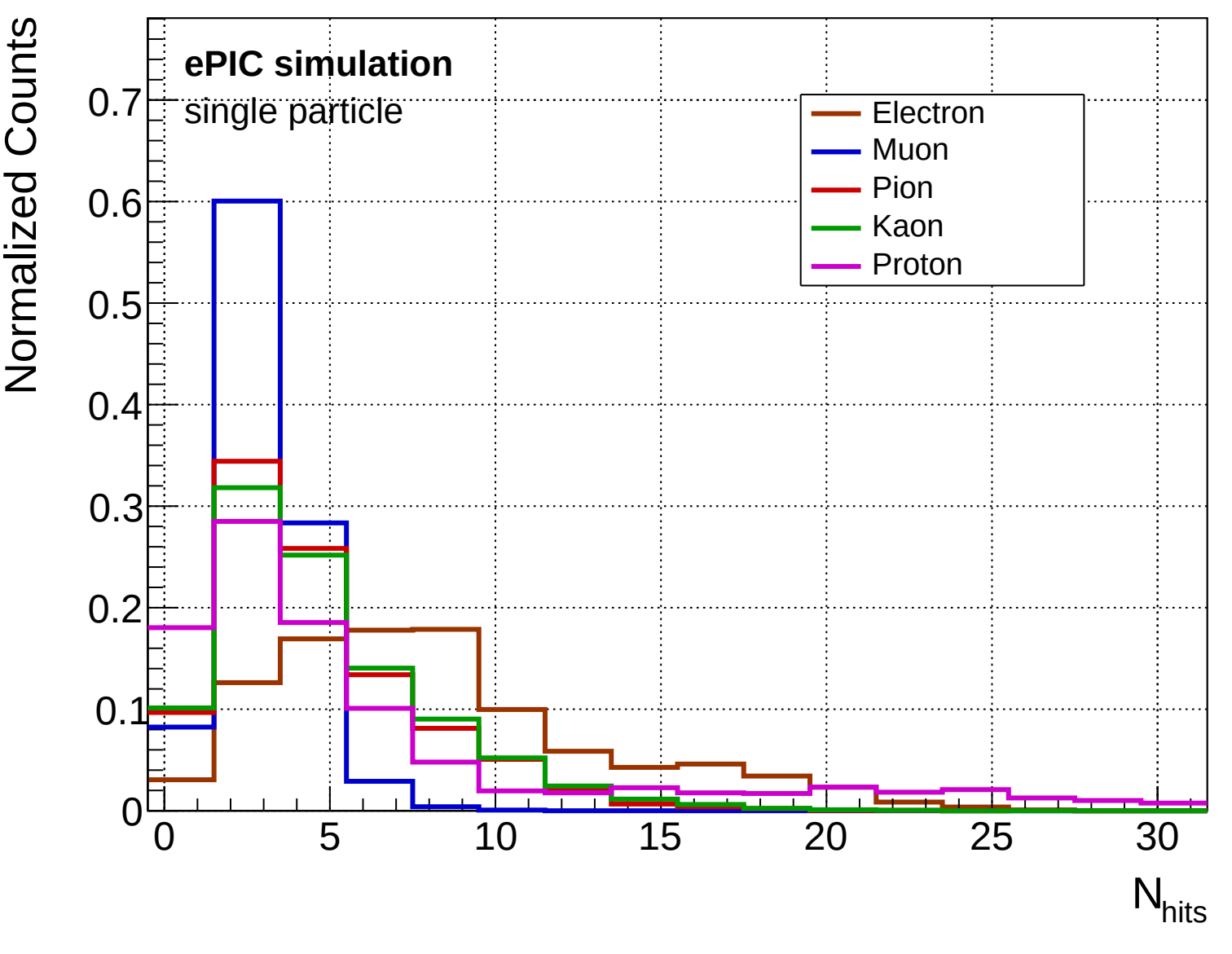

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
–0.8
–0.6
–0.4
–0.2
0
0.2
0.4
0.6
0.8
Δη

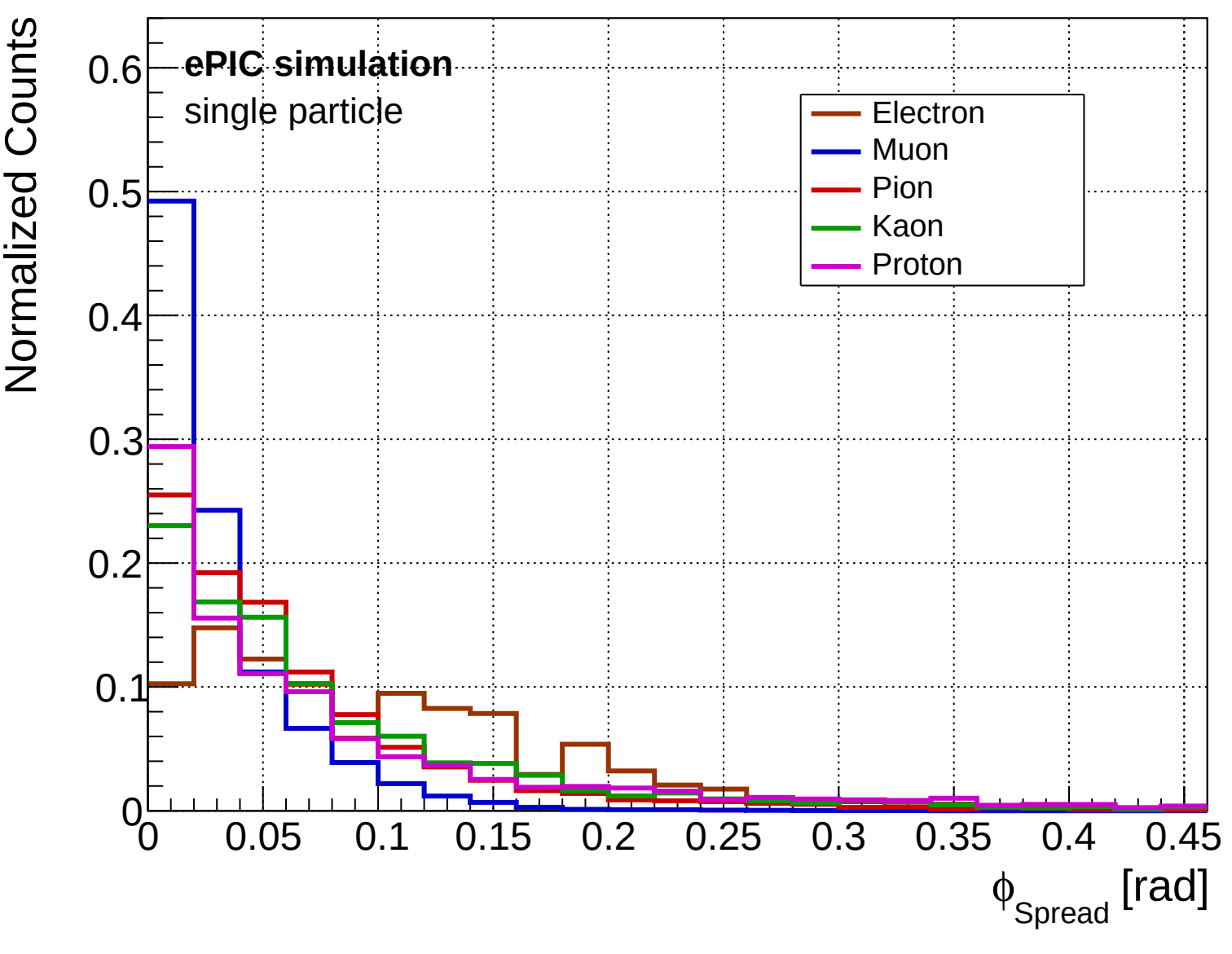

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.025
0.02
0.015
0.01
0.005
0
–1
–0.8
–0.6
–0.4
–0.2
0
0.2
0.4
0.6
0.8
Δϕ

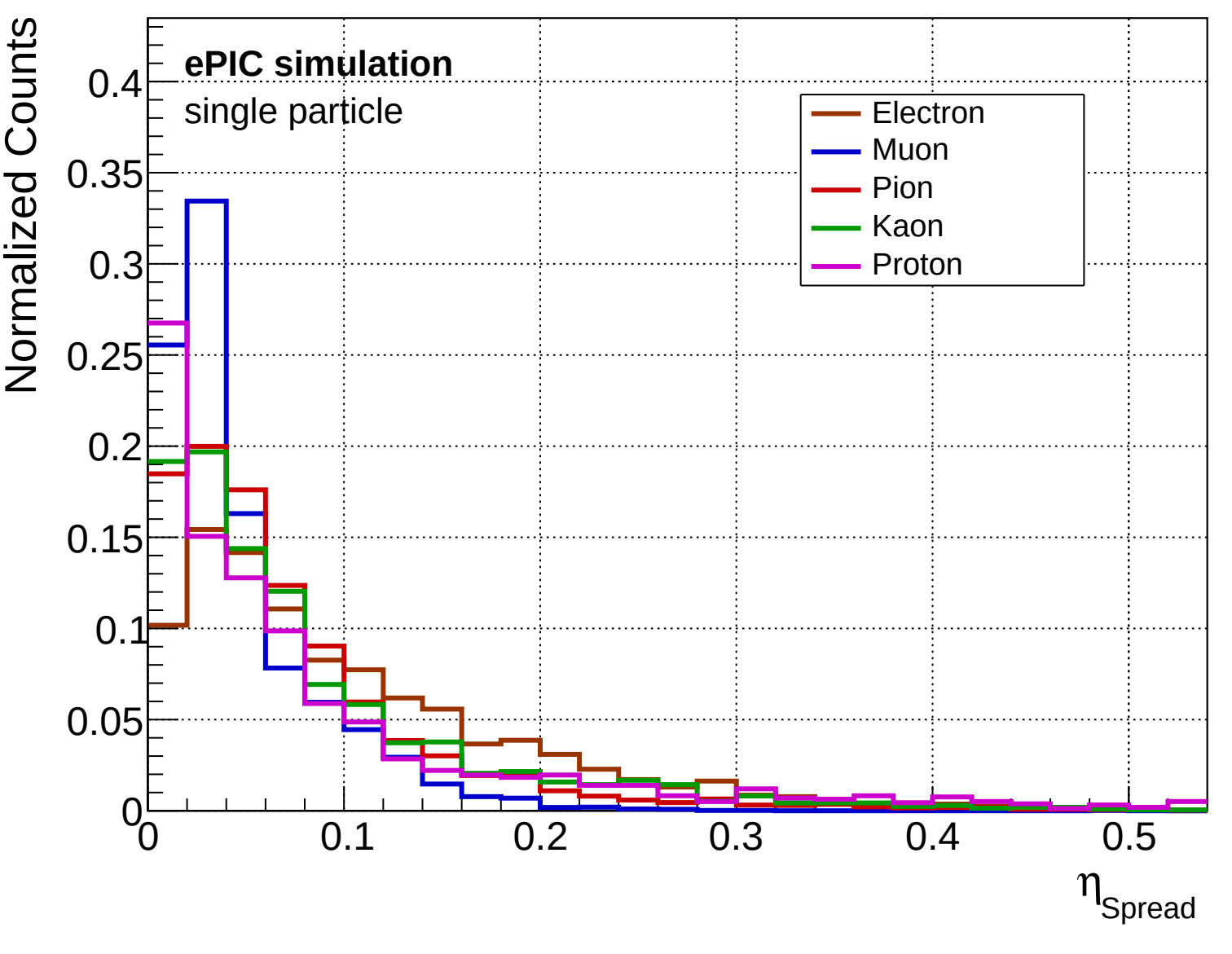

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.05
0.1
0.15
0.2
0.25
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
E_calo [GeV]

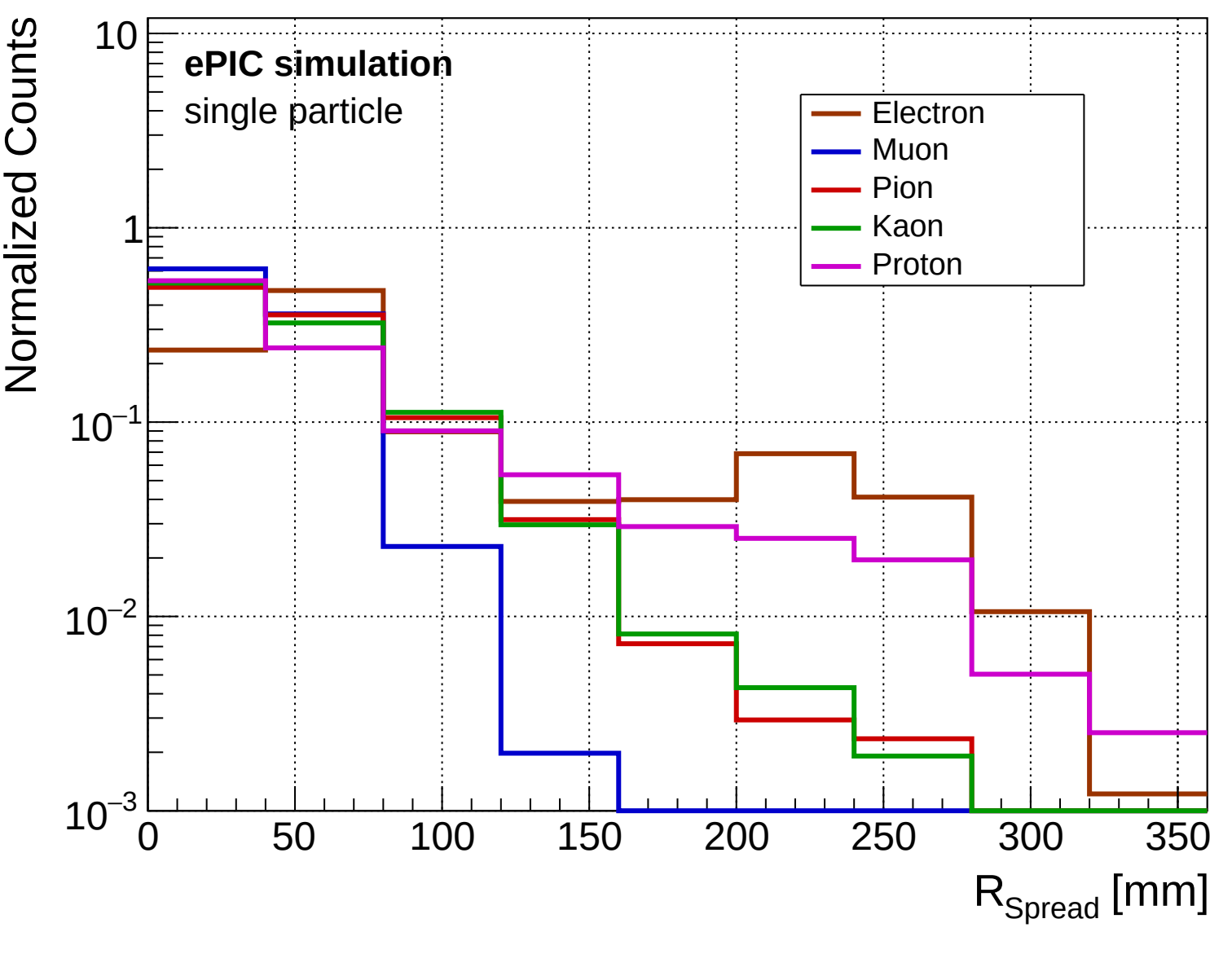

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
E/p
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2

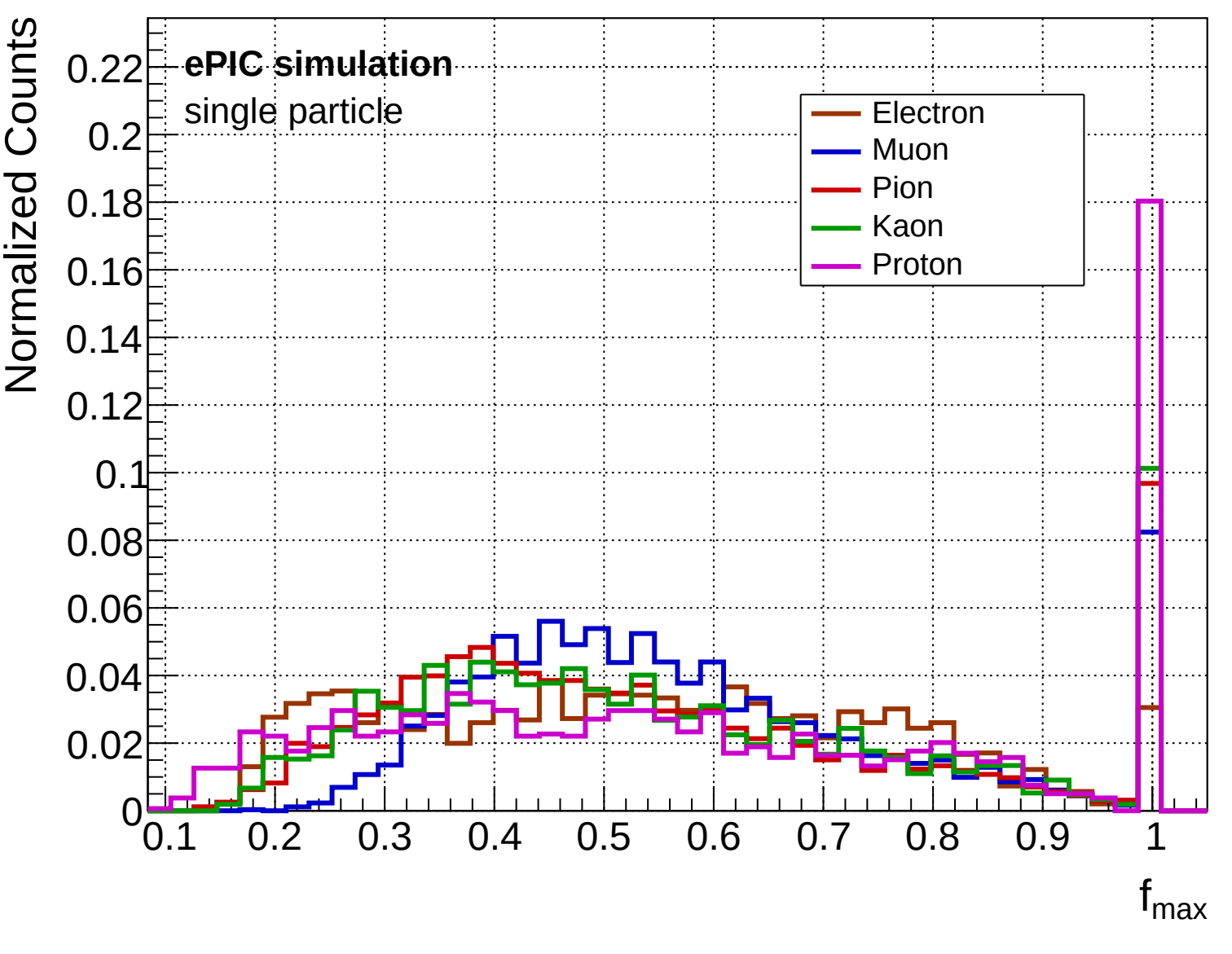

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
E_avg [GeV]

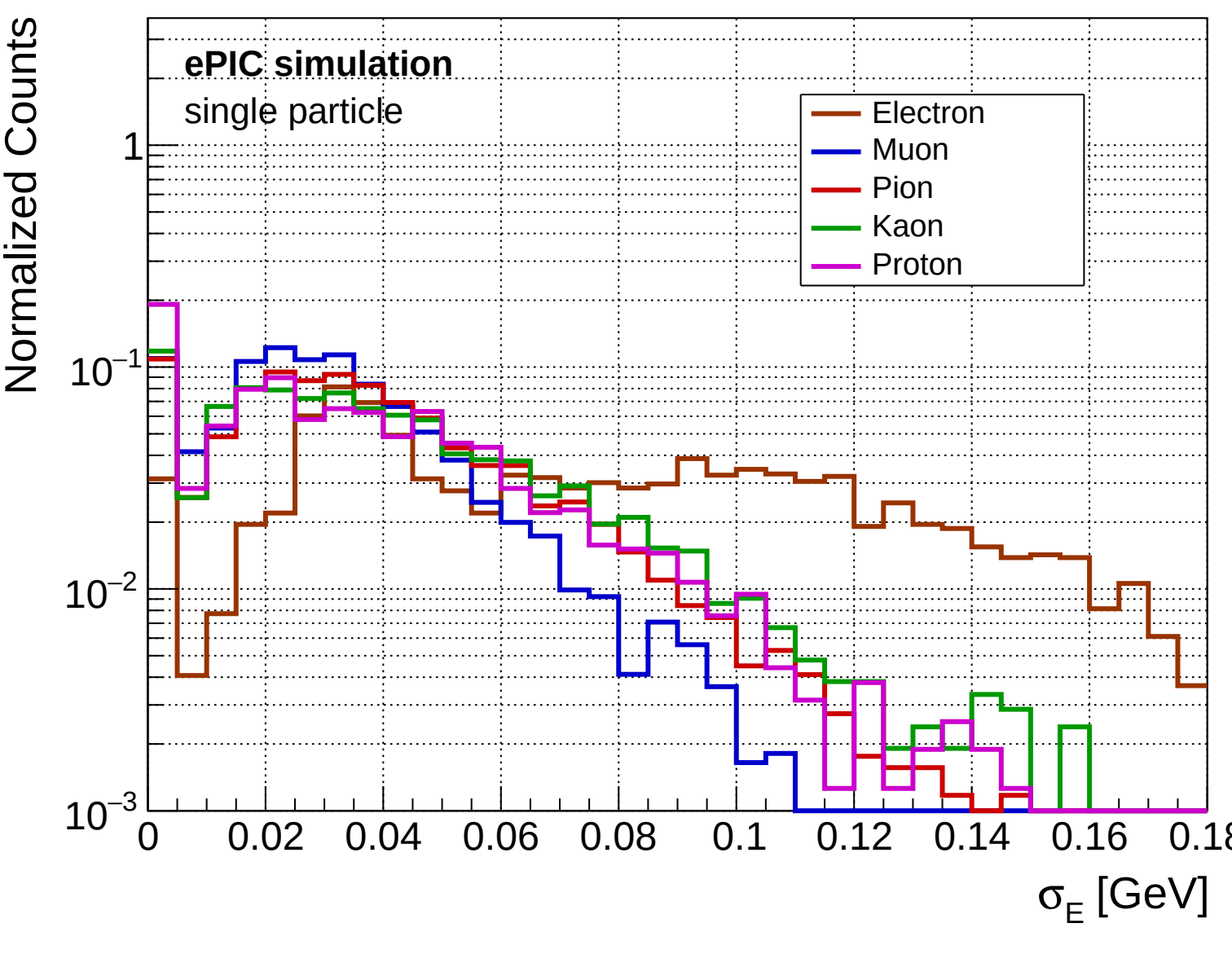

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
5
10
15
20
25
30
N_hits

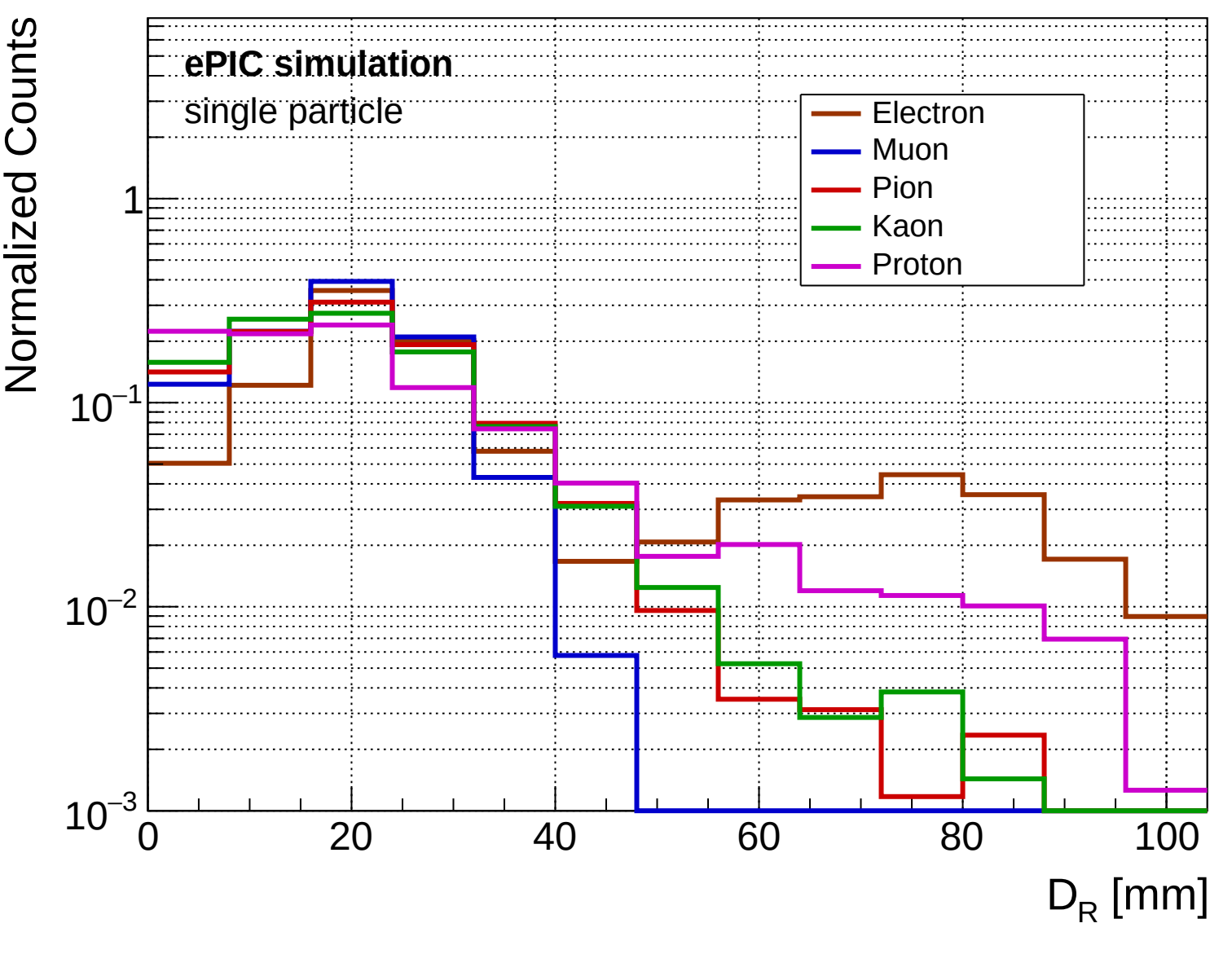

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.1
0.2
0.3
0.4
0.5
0.6
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
ϕ_{Spread} [rad]

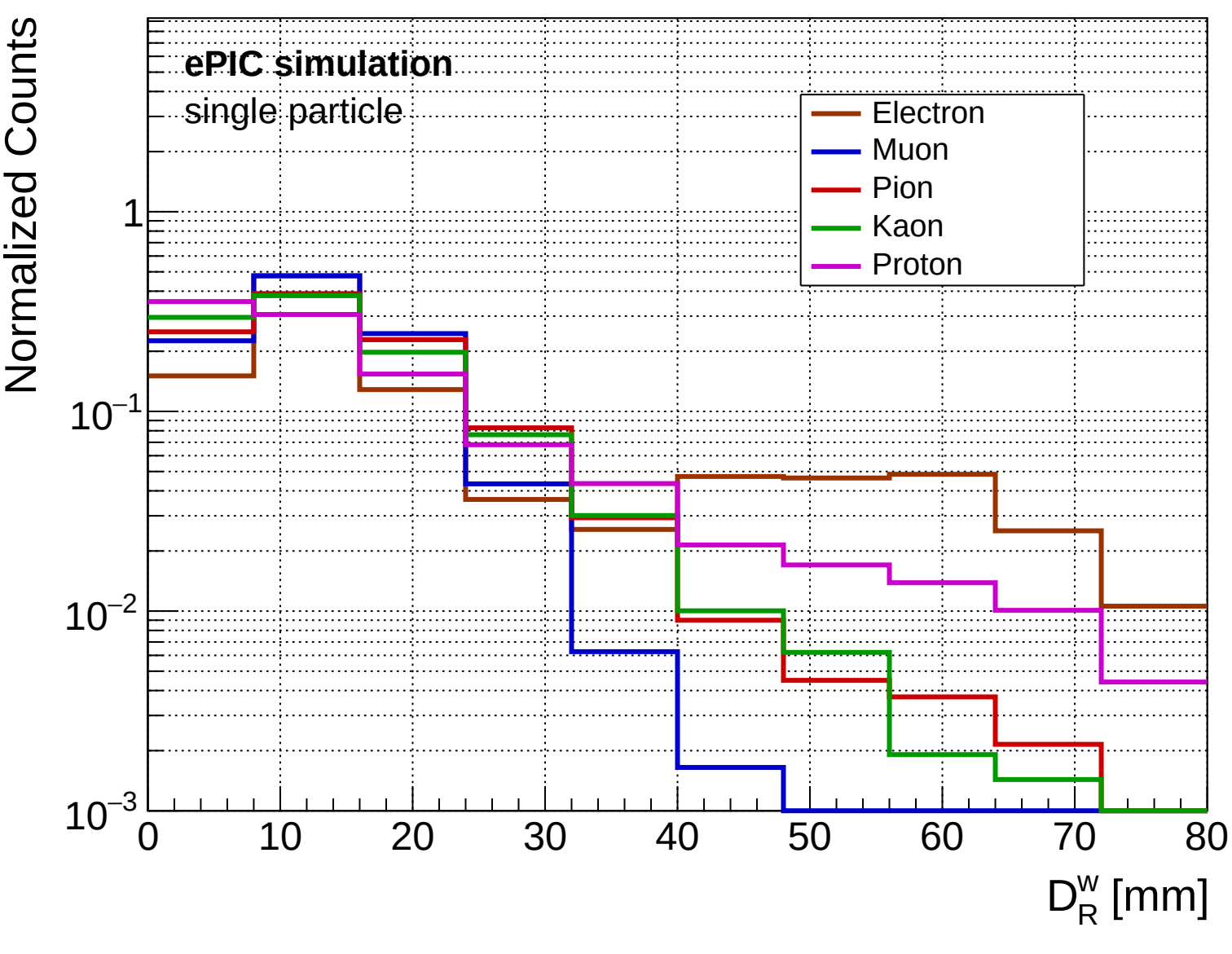

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0
0.1
0.2
0.3
0.4
0.5
η_{Spread}

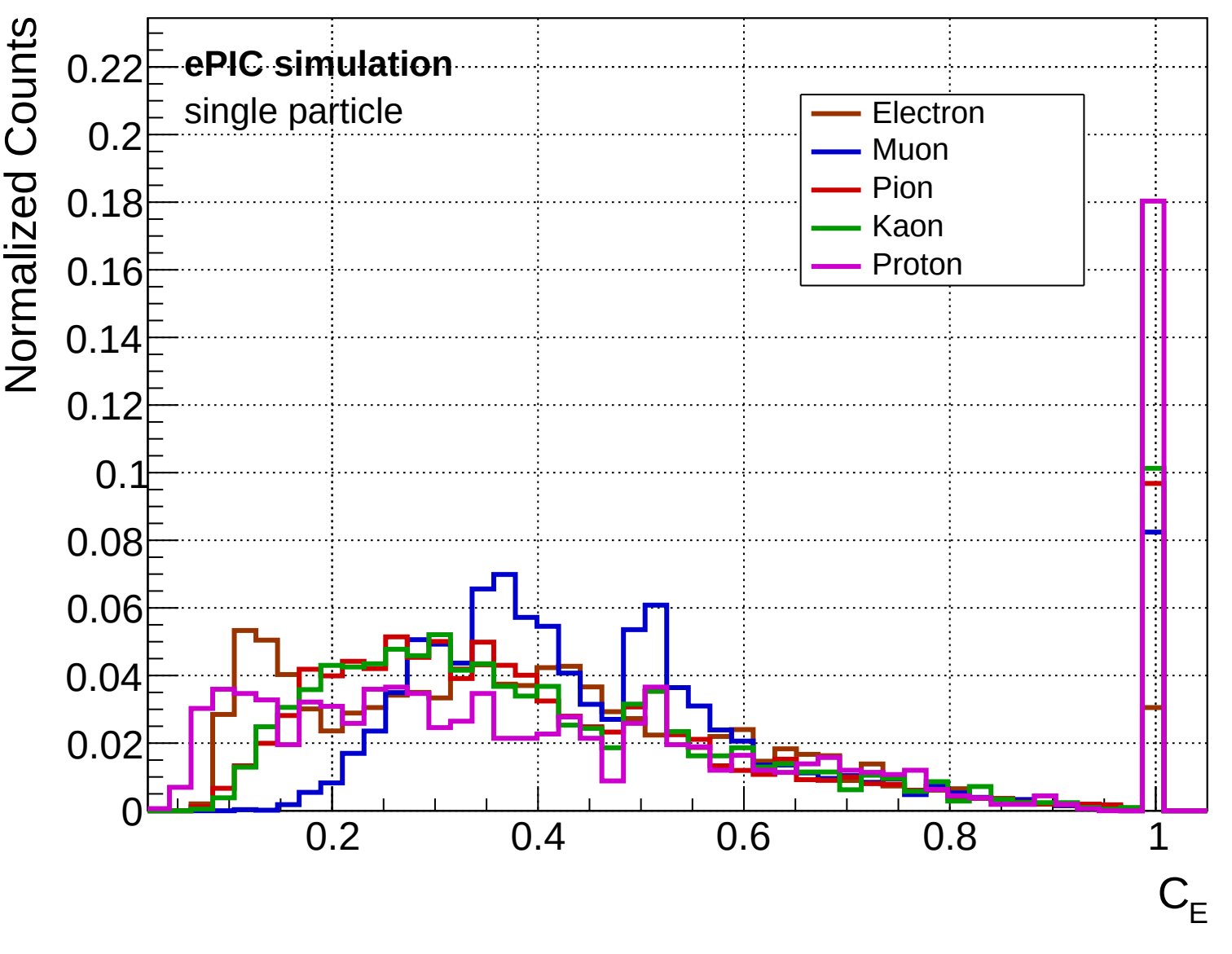

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
10
1
10^{-1}
10^{-2}
10^{-3}
0
50
100
150
200
250
300
350
R_{Spread} [mm]

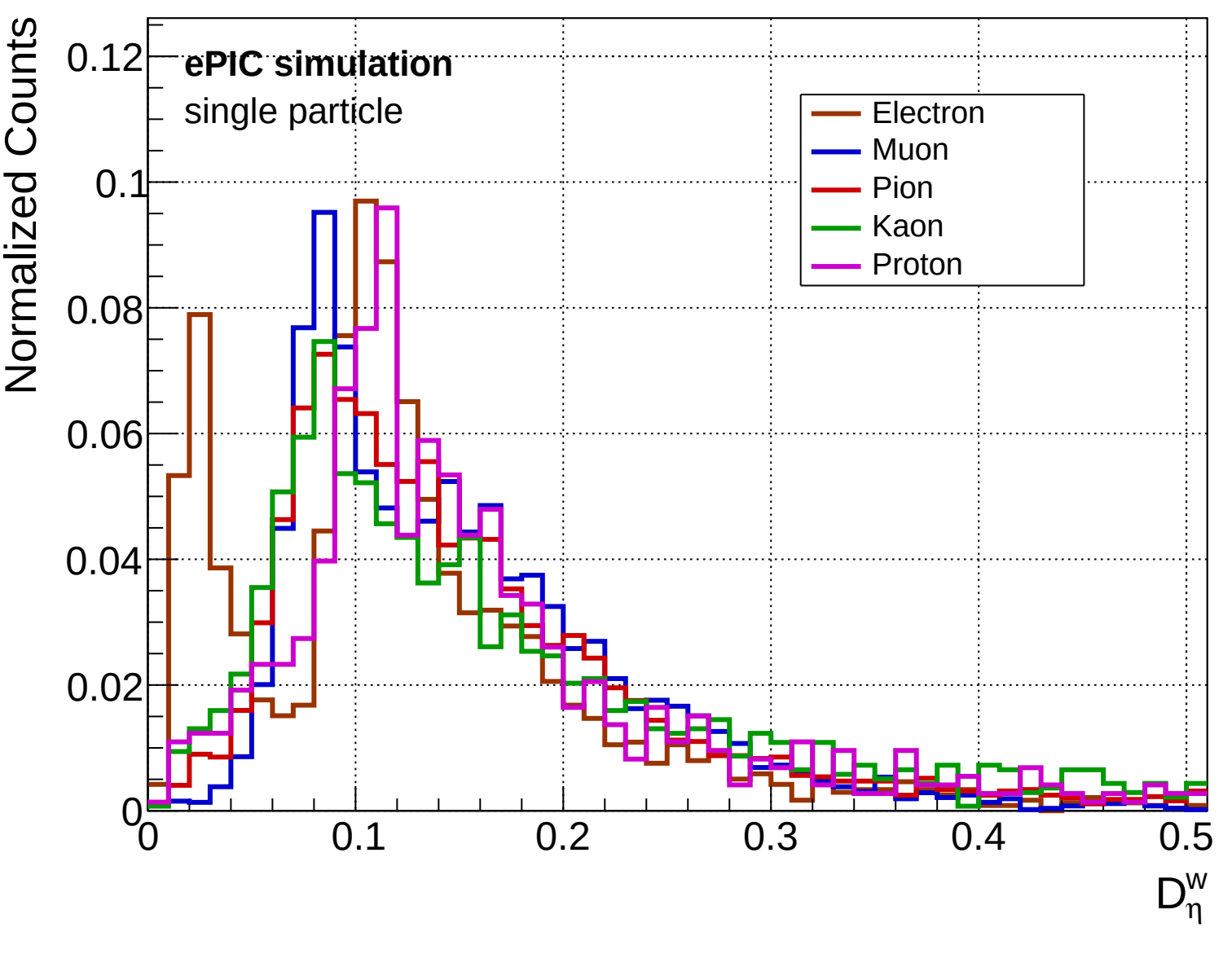

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
f_{max}

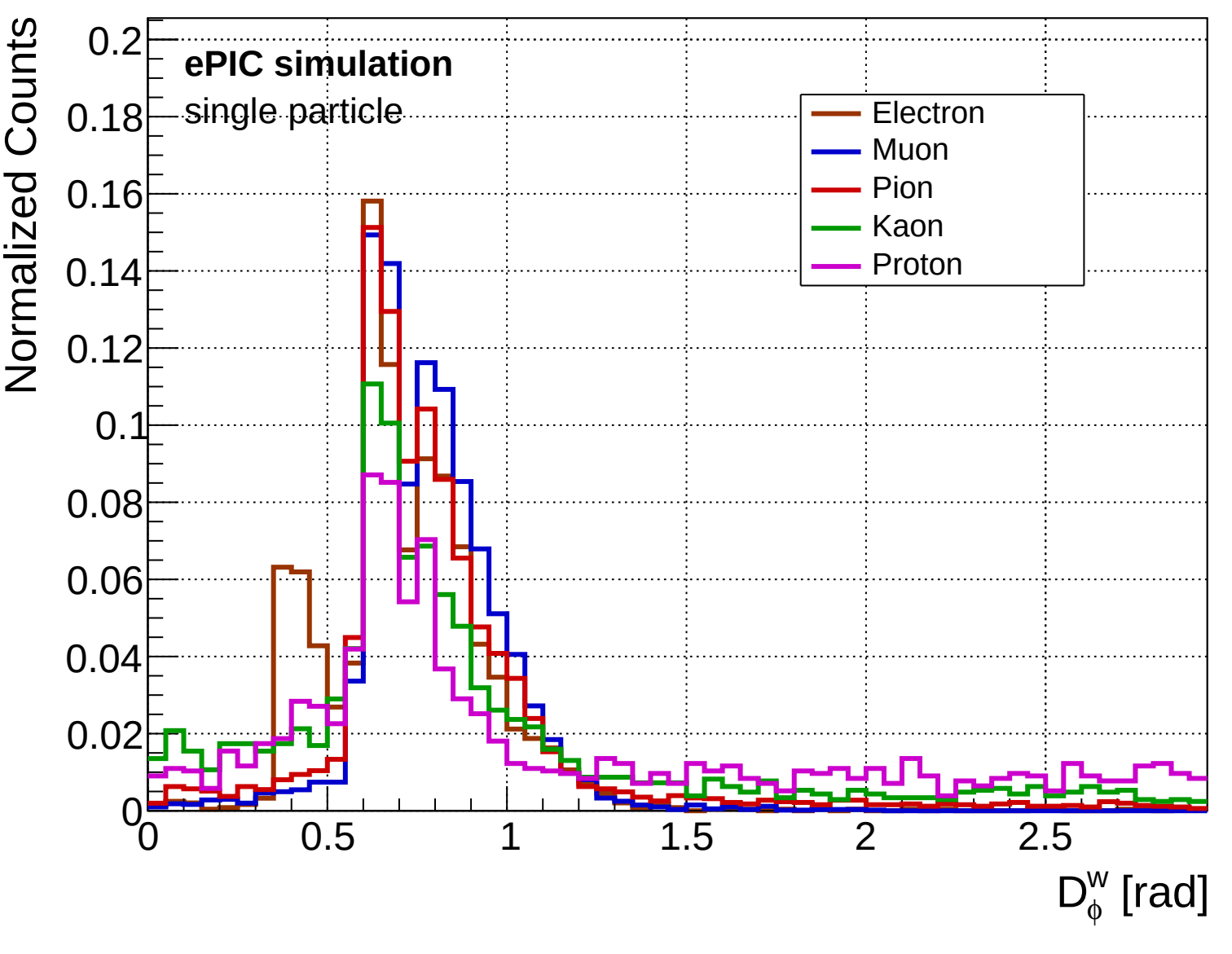

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
σ_E [GeV]

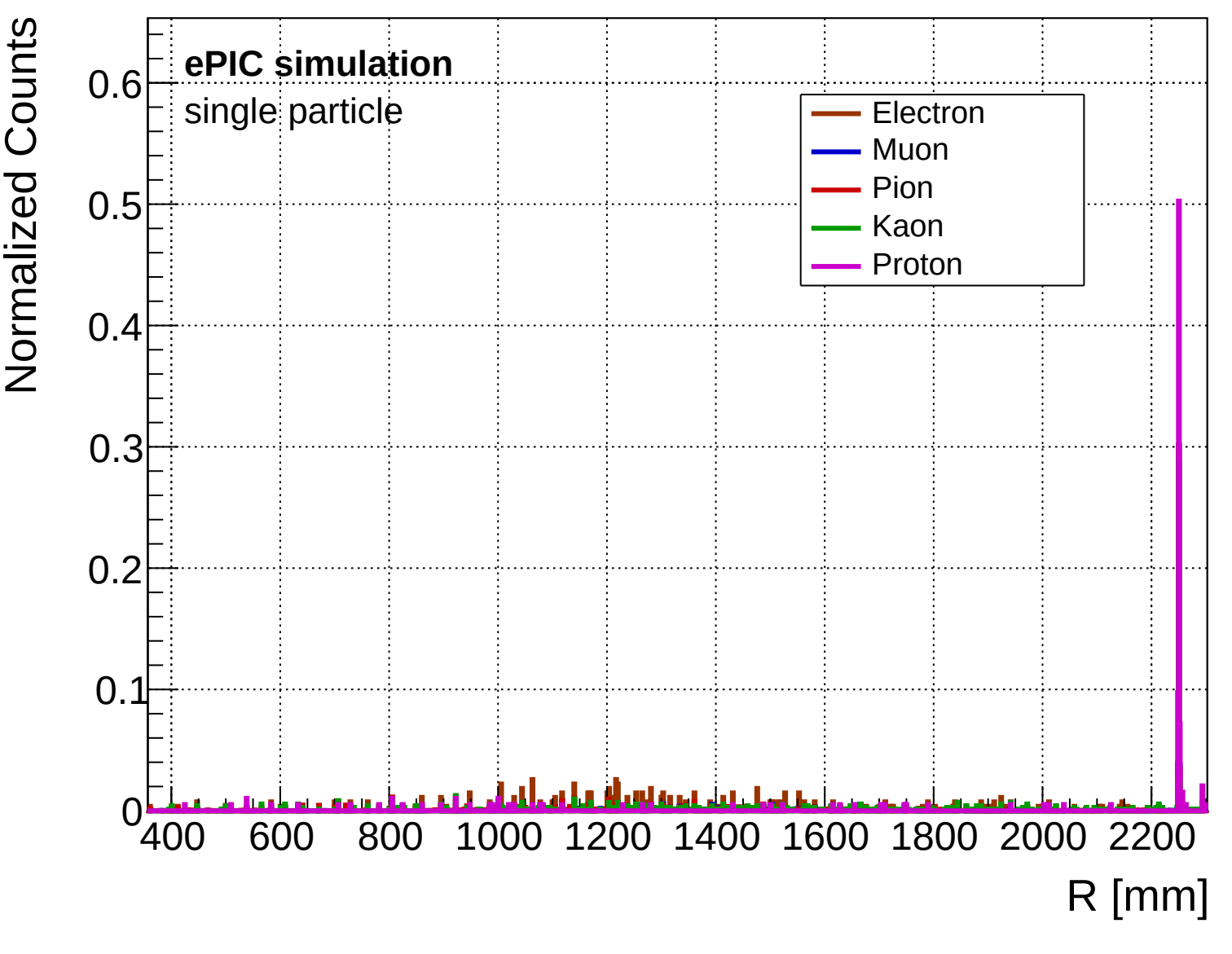

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
20
40
60
80
100
D_R [mm]

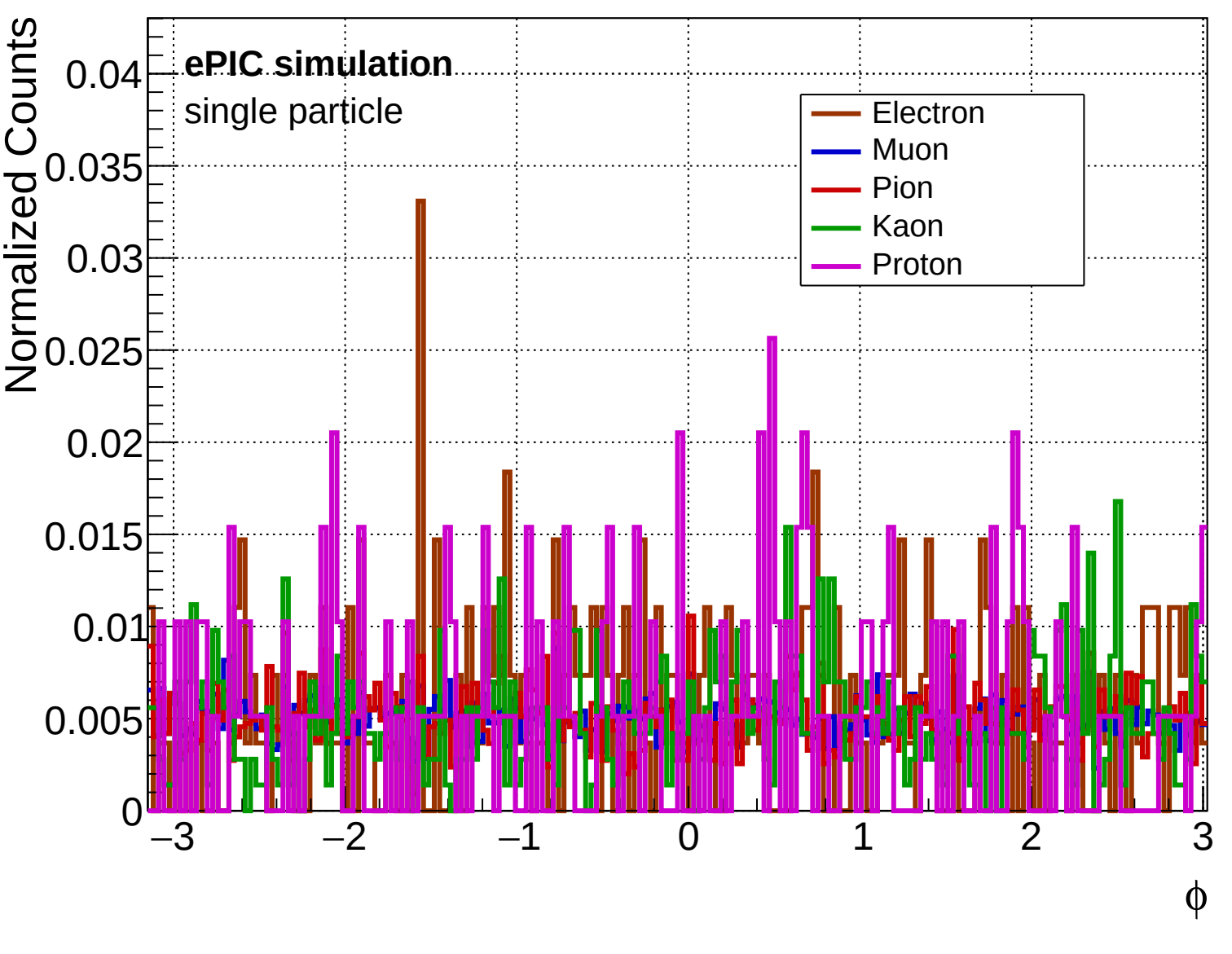

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
10
20
30
40
50
60
70
80
D_R^W [mm]

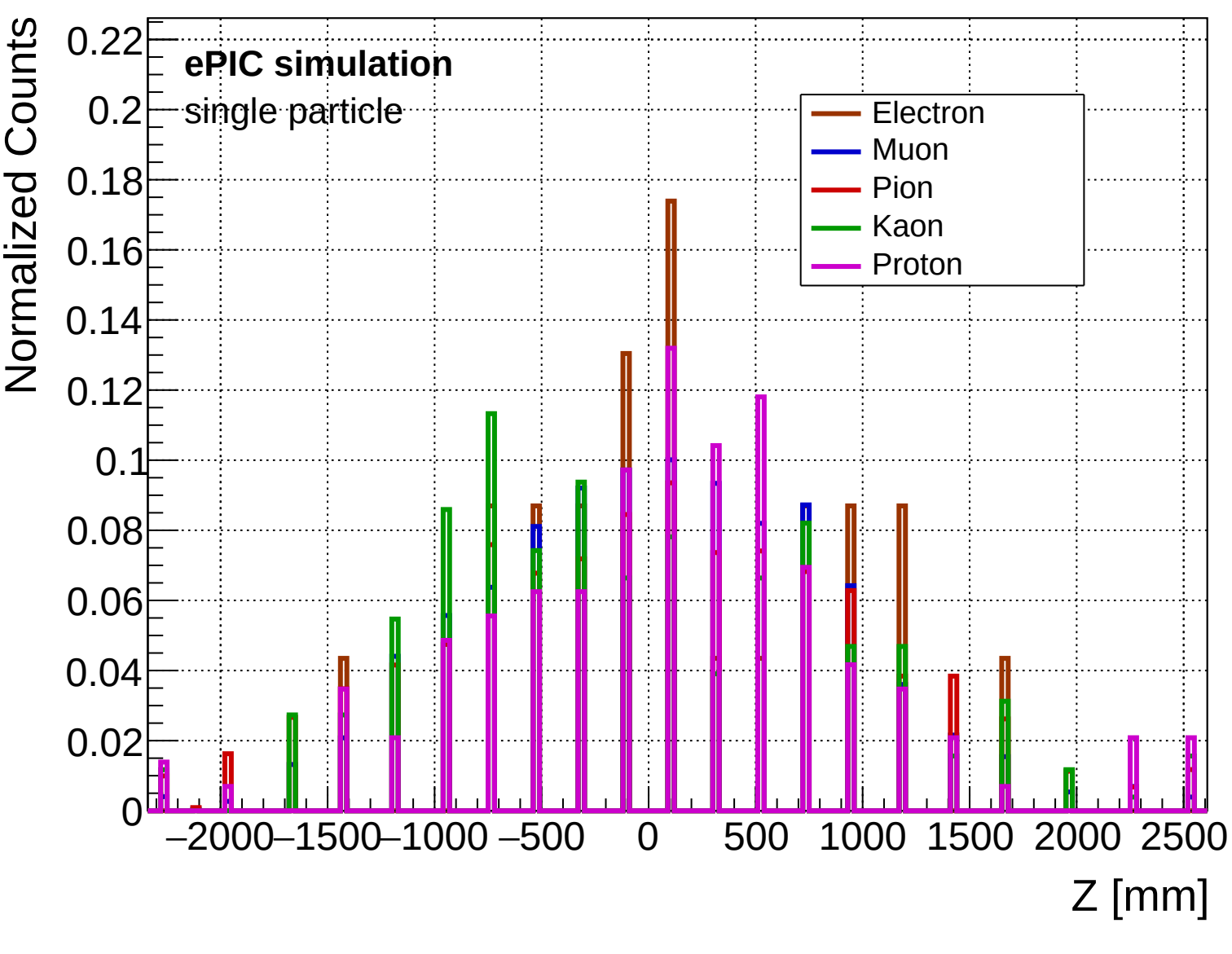

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0.2
0.4
0.6
0.8
1
C_E

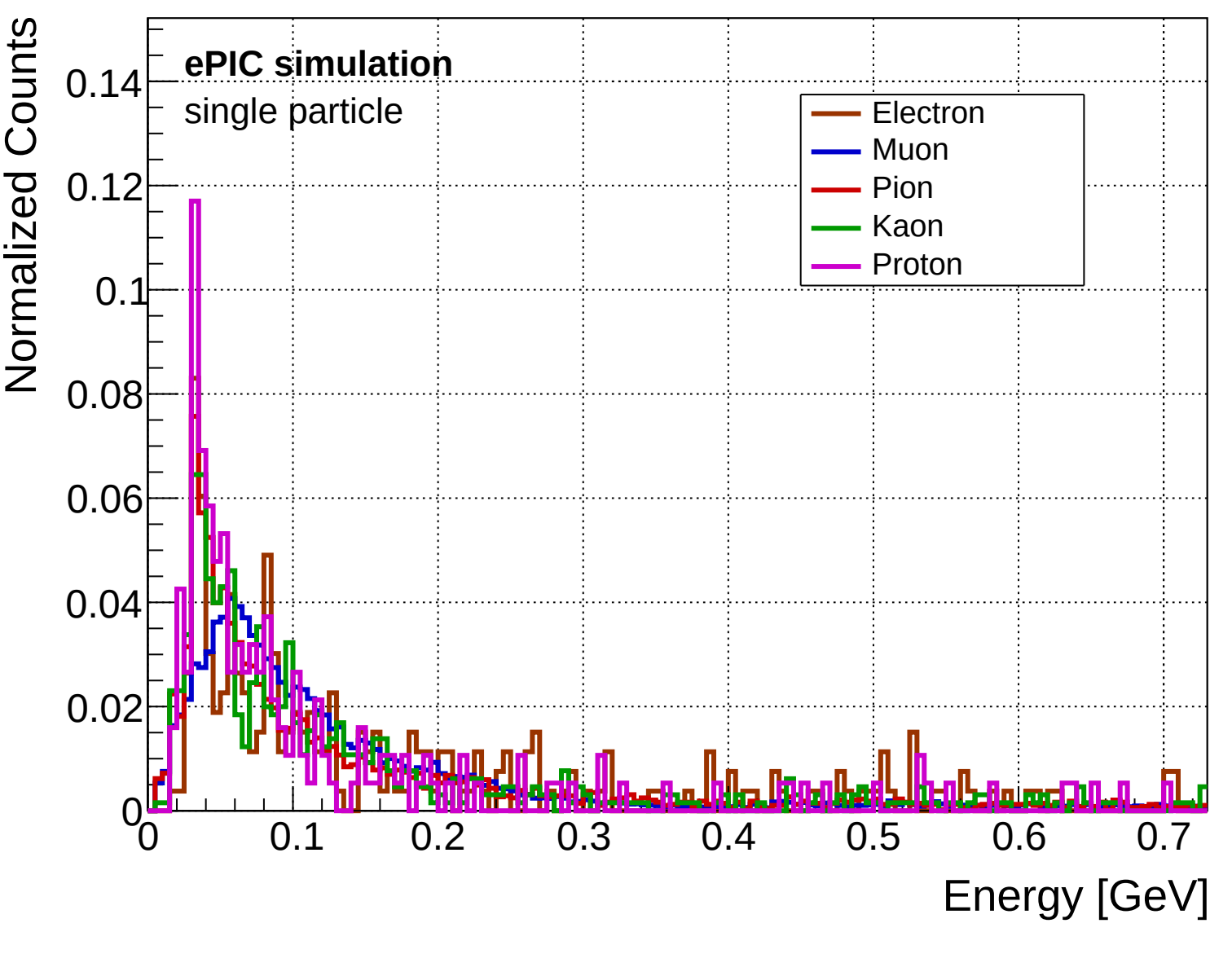

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.1
0.2
0.3
0.4
0.5
D_{η}^{w}

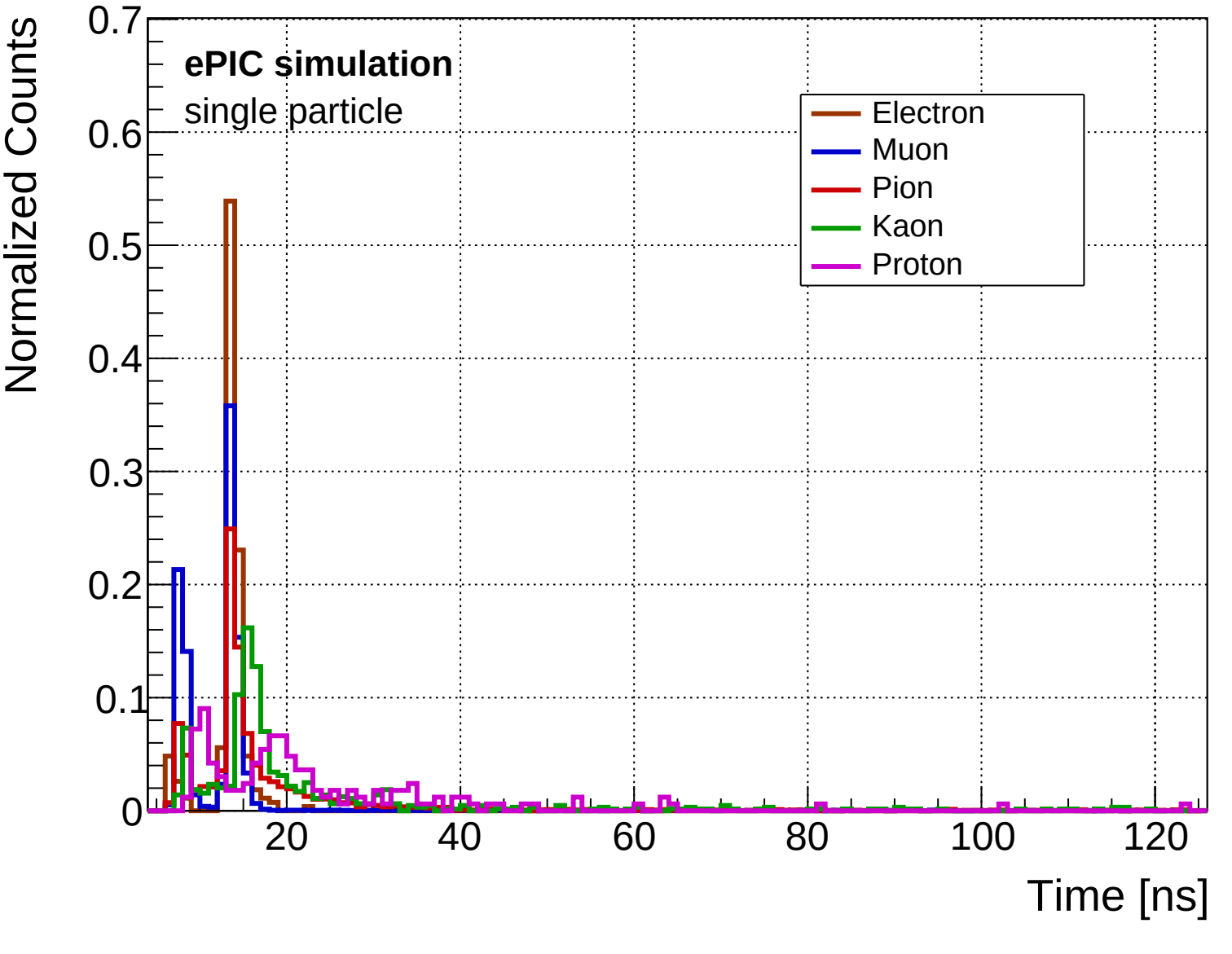

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.5
1
1.5
2
2.5
D_{ϕ}^{w} [rad]

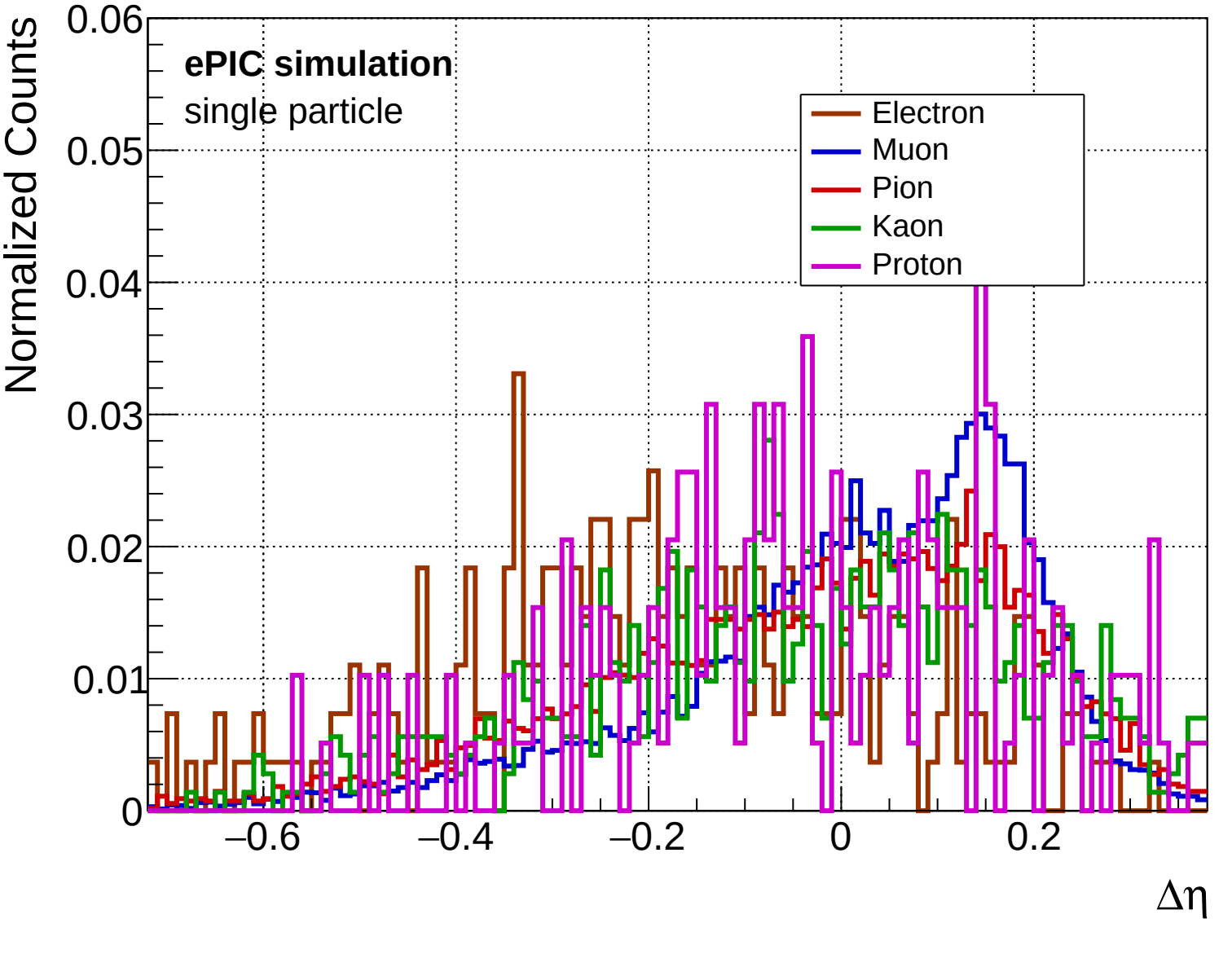

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.1
0.2
0.3
0.4
0.5
0.6
400
600
800
1000
1200
1400
1600
1800
2000
2200
R [mm]

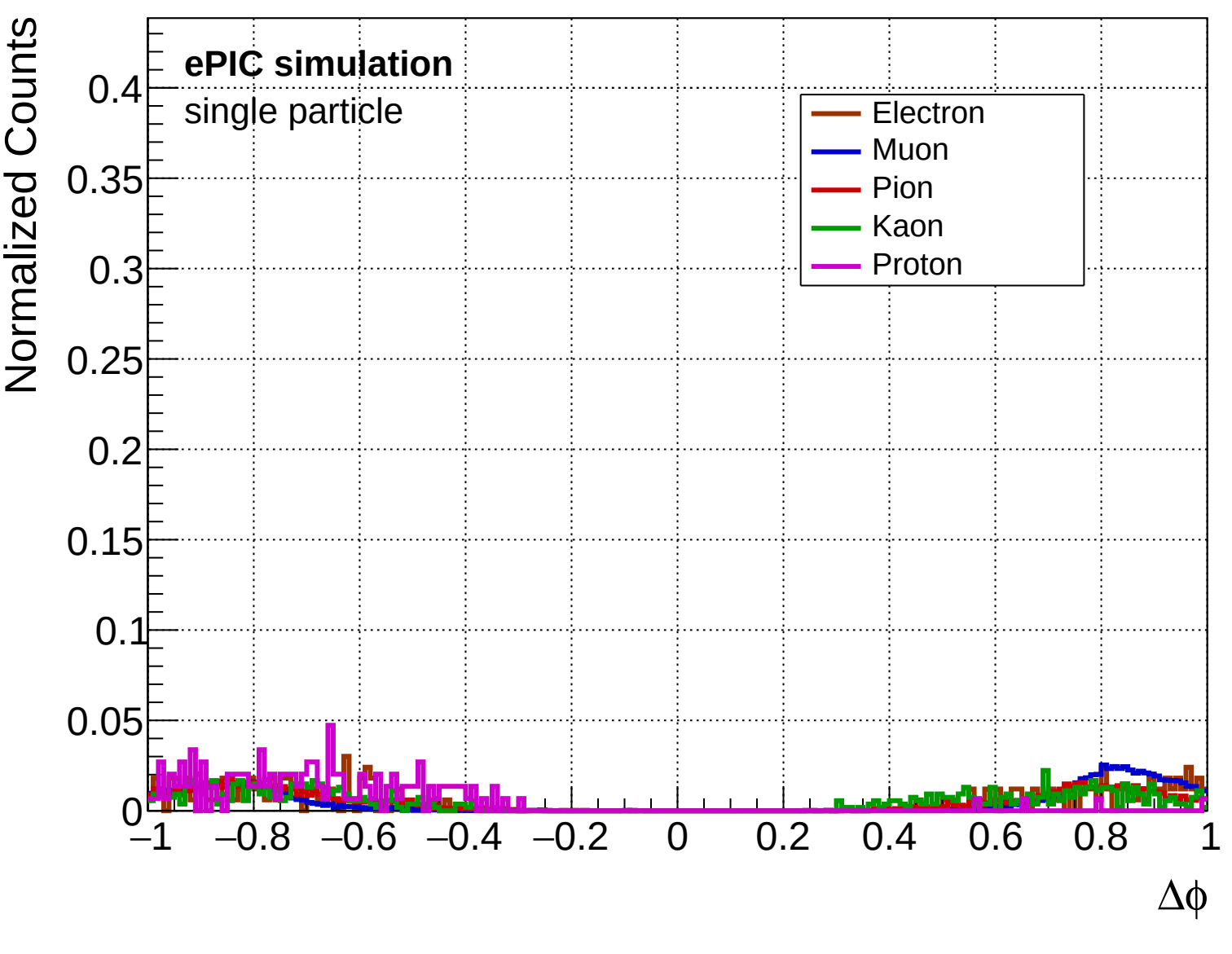

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−3
−2
−1
0
1
2
3
ϕ

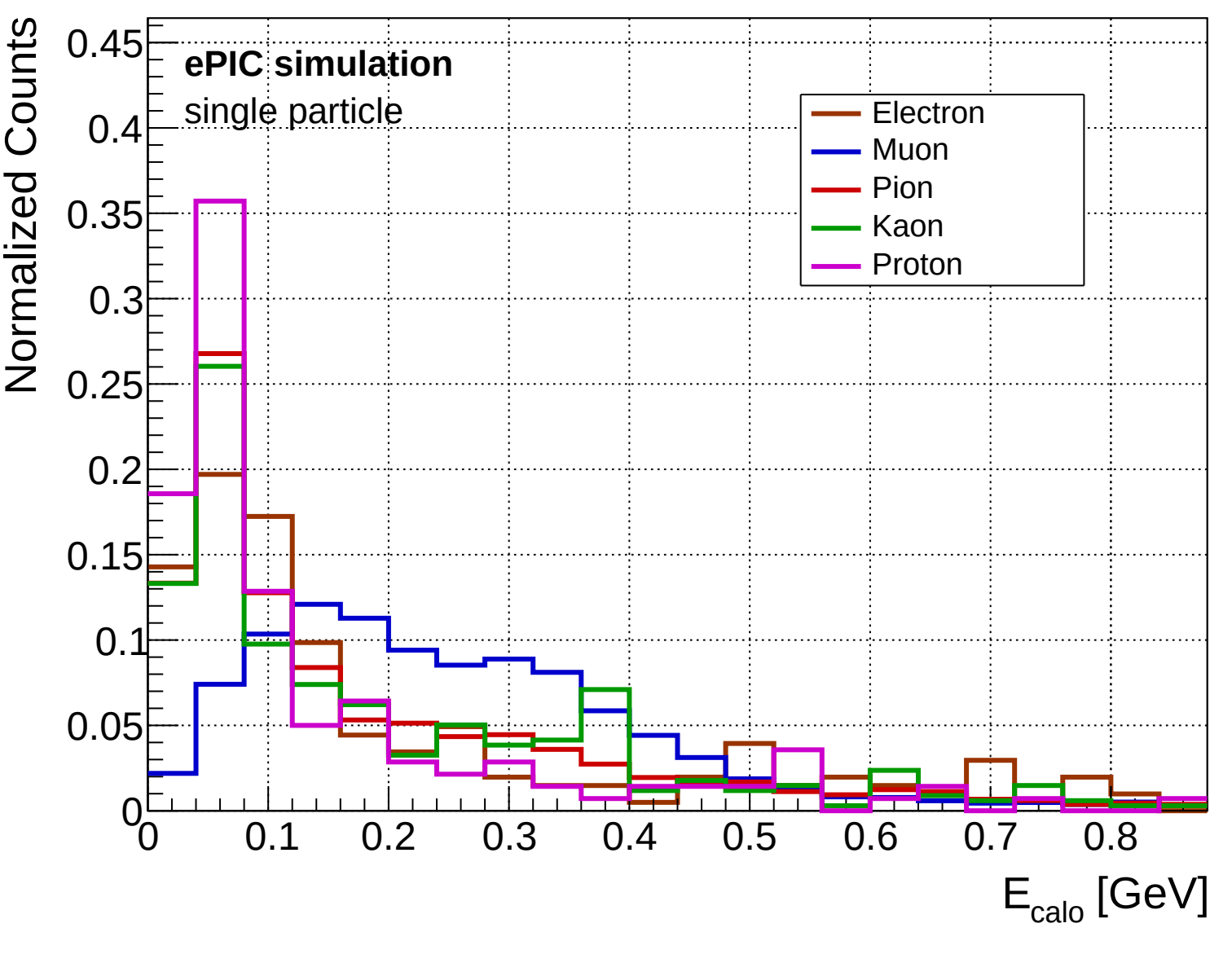

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
–2000
–1500
–1000
–500
0
500
1000
1500
2000
2500
Z [mm]

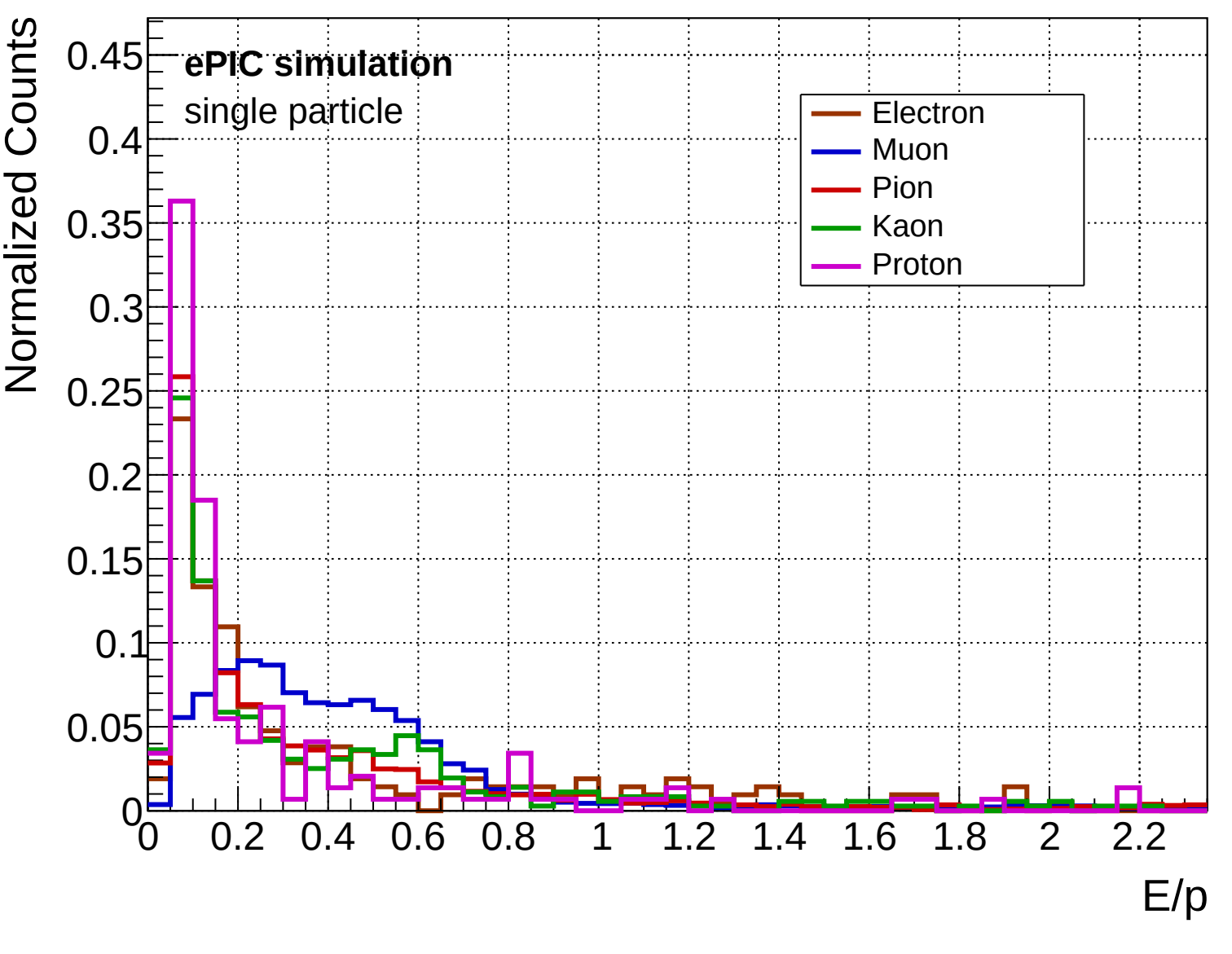

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Energy [GeV]

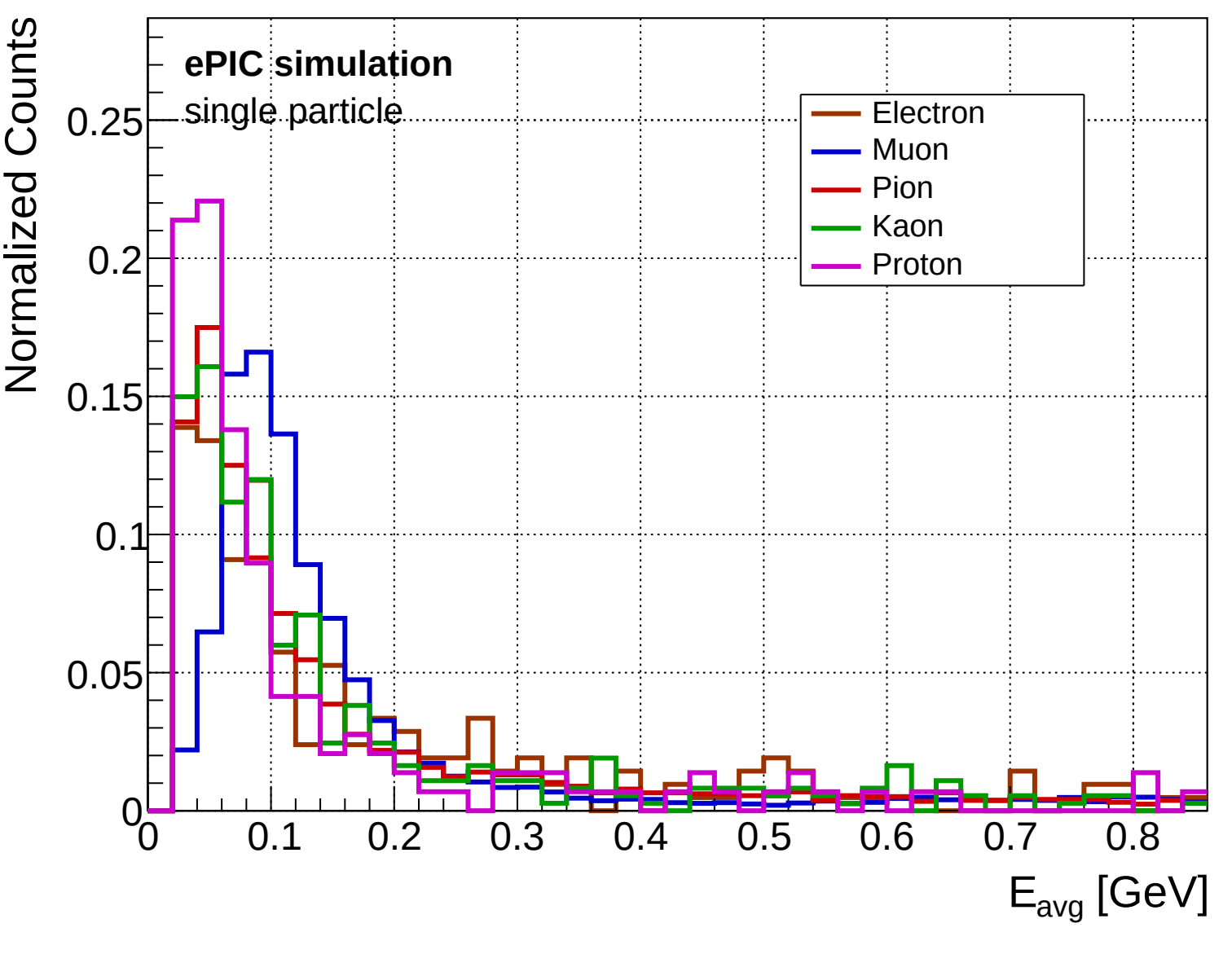

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
100
120
Time [ns]

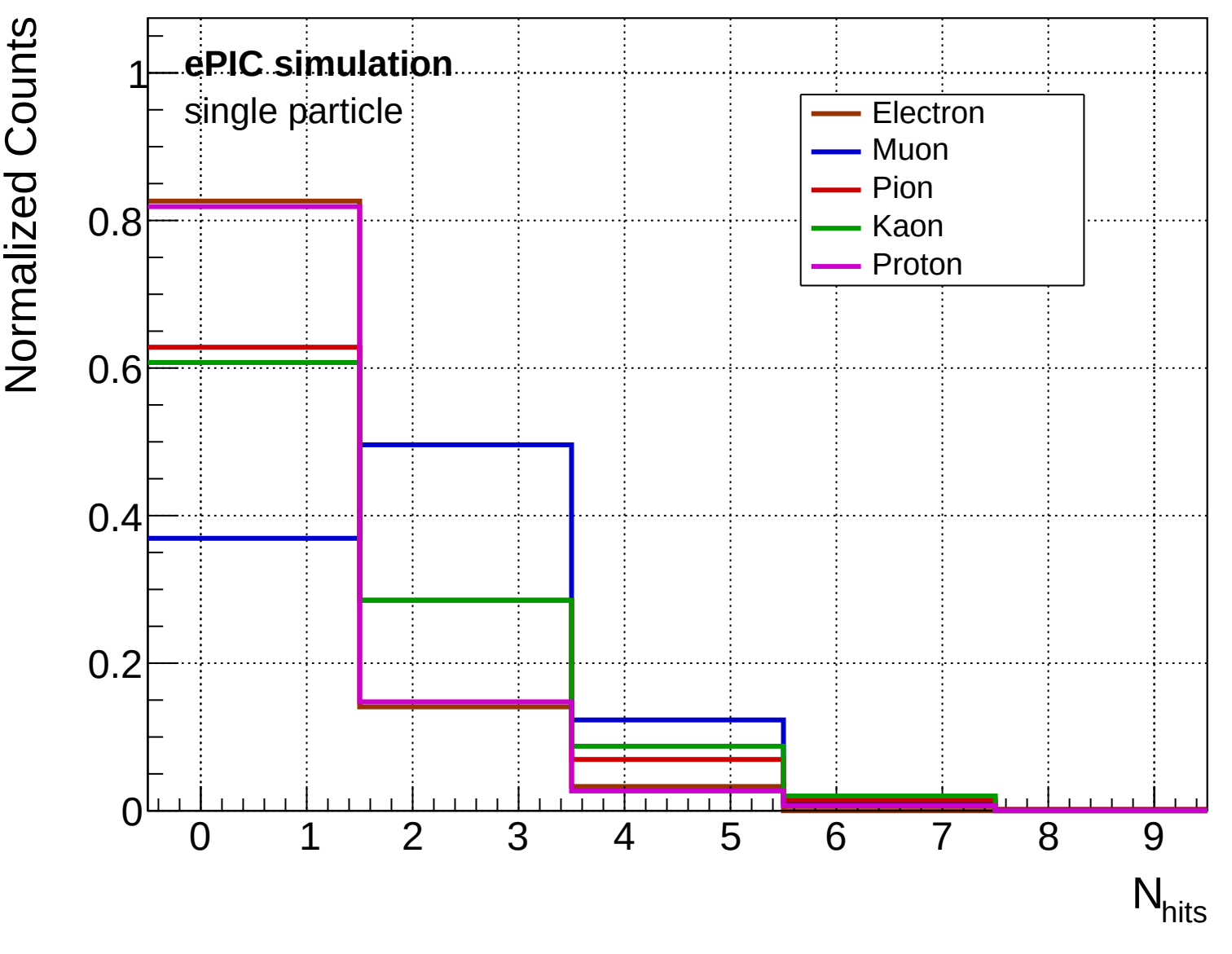

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
–0.6
–0.4
–0.2
0
0.2
Δη

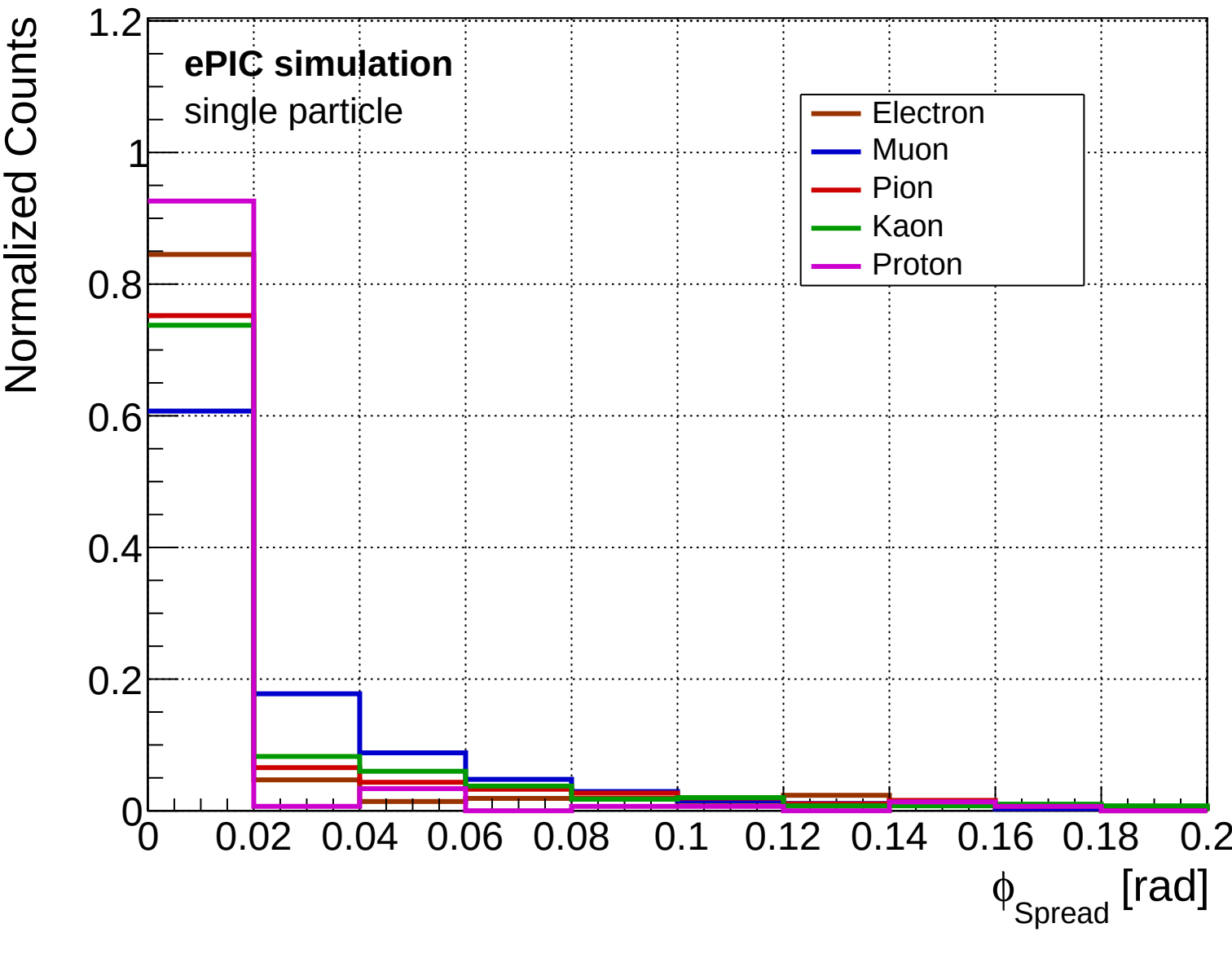

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
–1
–0.8
–0.6
–0.4
–0.2
0
0.2
0.4
0.6
0.8
1
Δϕ

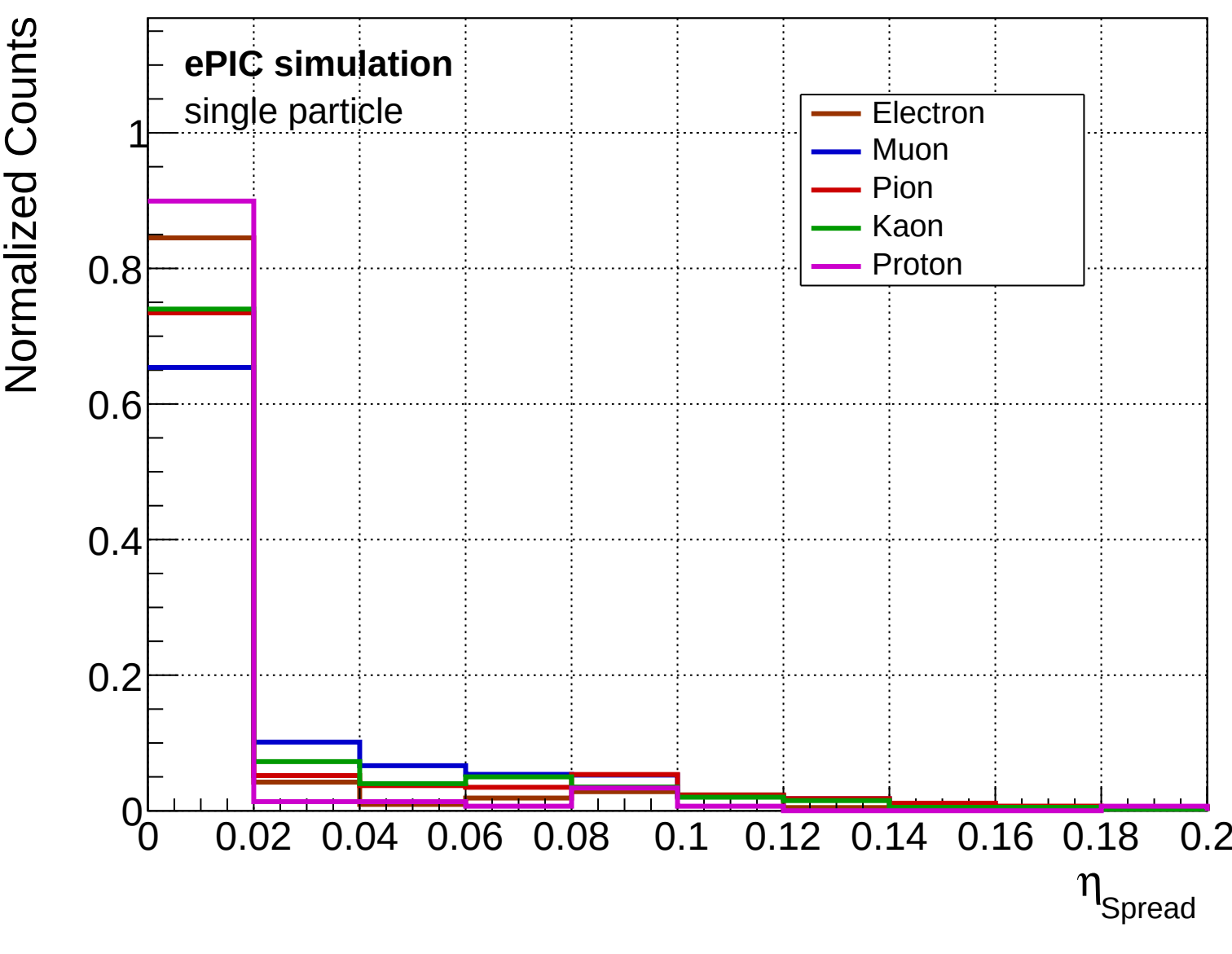

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
E_calo [GeV]

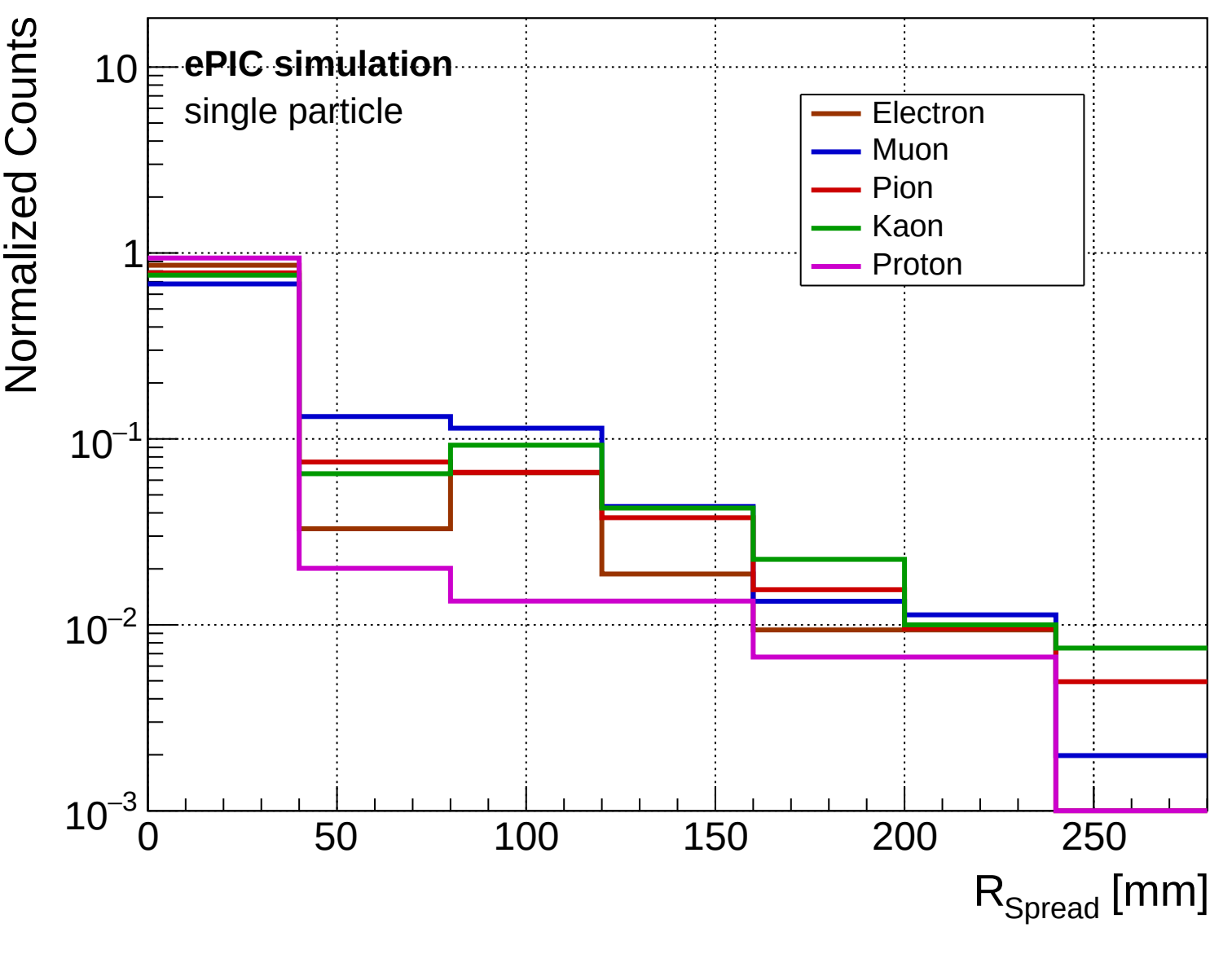

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
E/p
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2

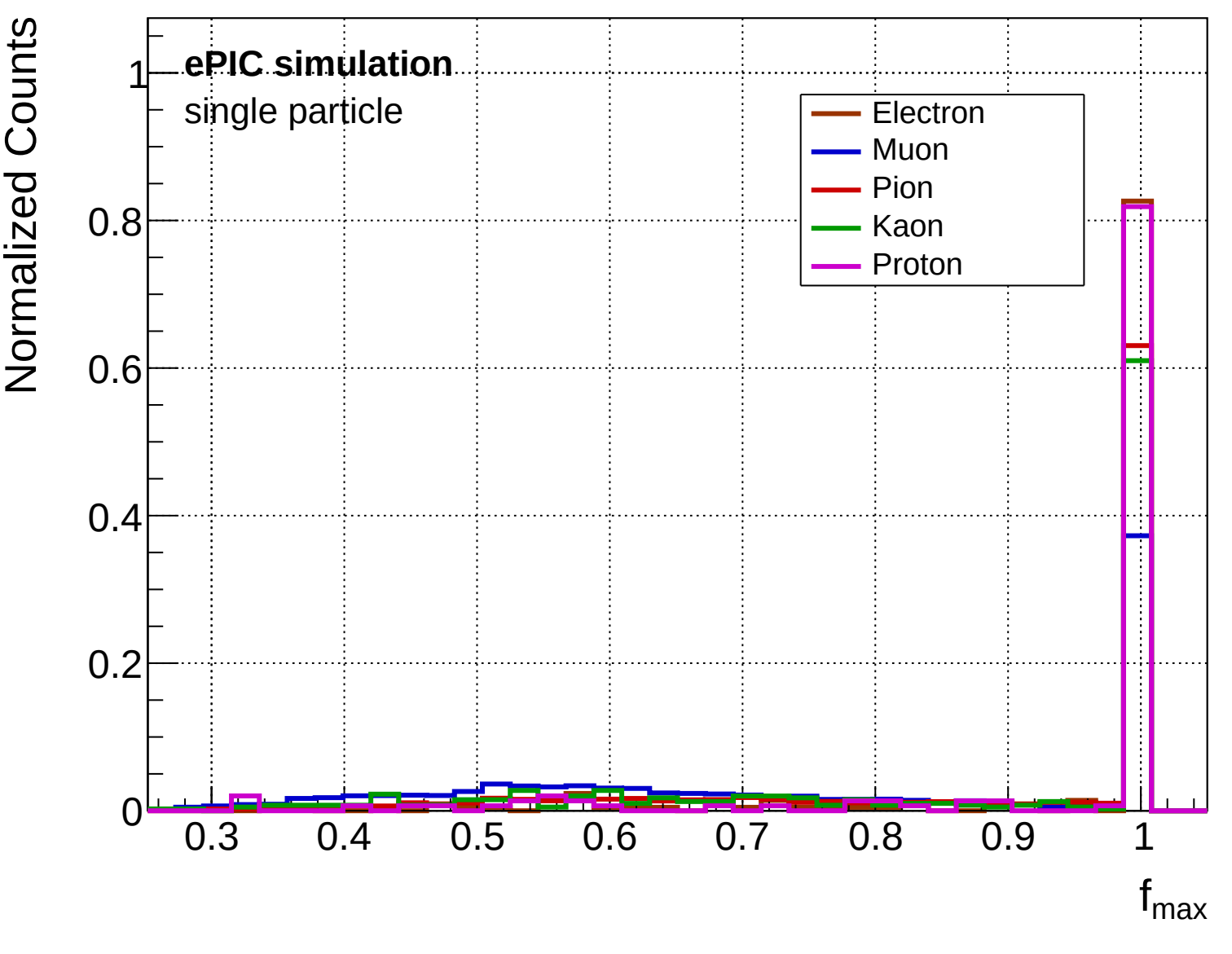

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.05
0.1
0.15
0.2
0.25
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
E_avg [GeV]

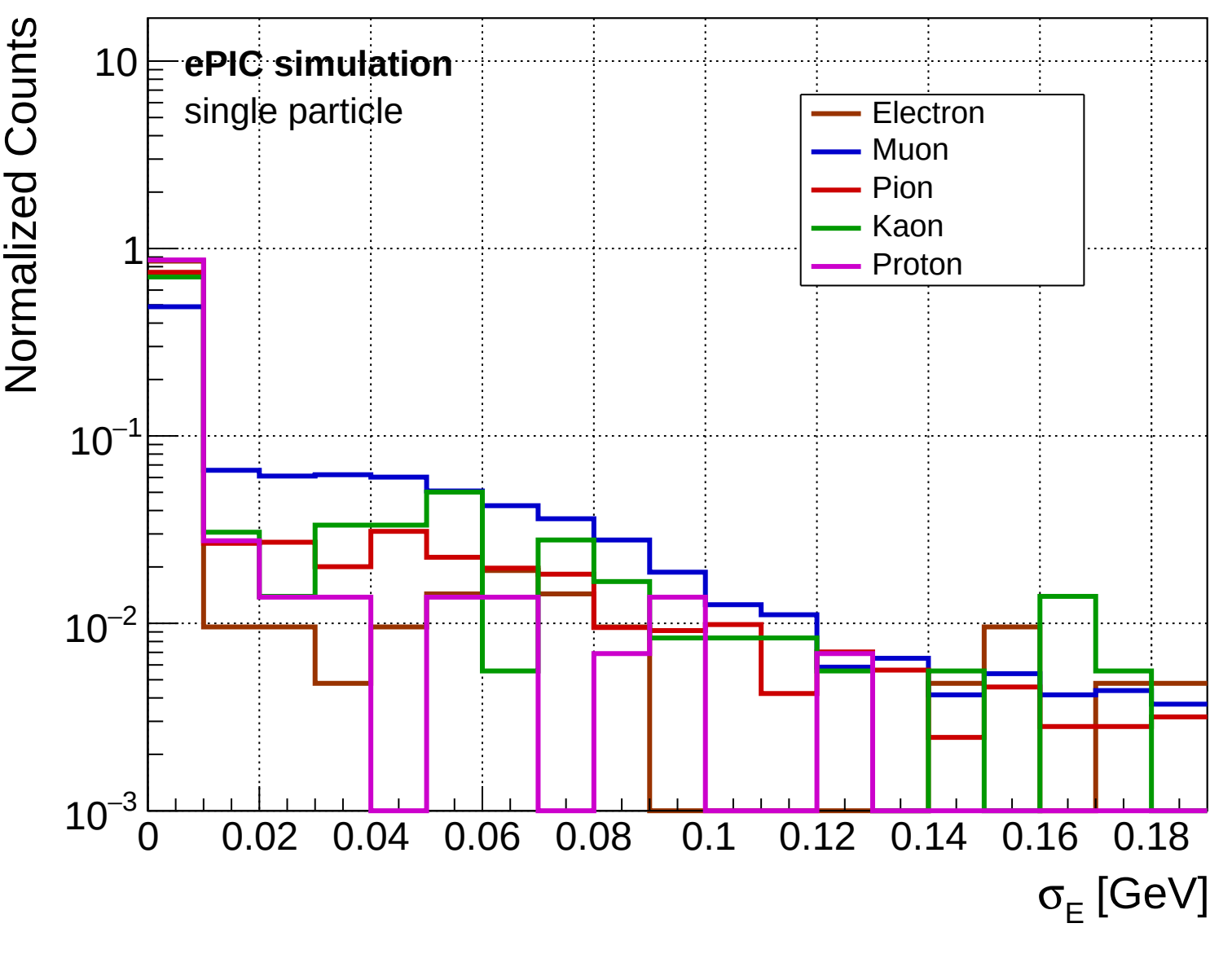

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.2
0.4
0.6
0.8
1
0
1
2
3
4
5
6
7
8
9
N_hits

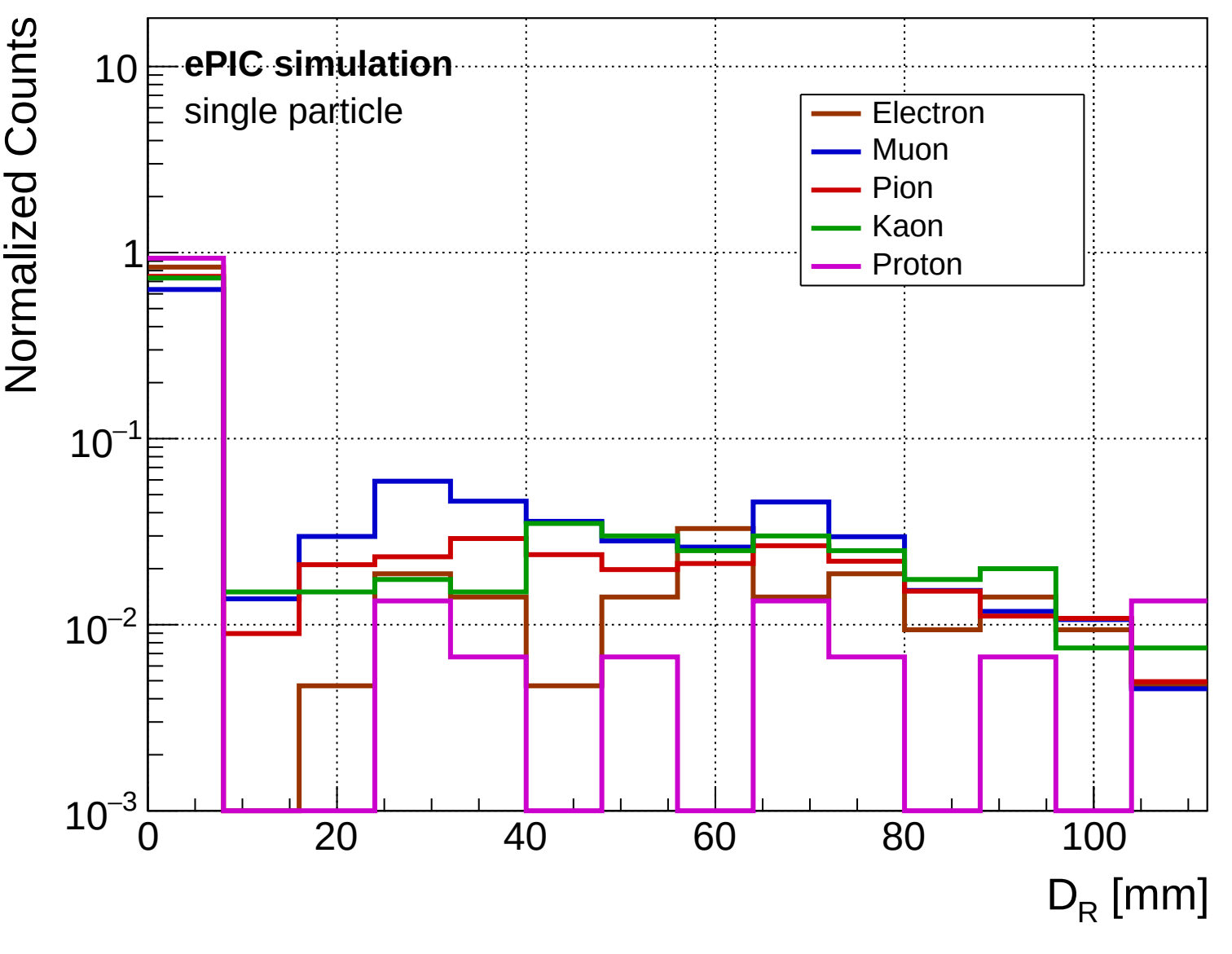

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
1.2
1
0.8
0.6
0.4
0.2
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
ϕ_{Spread} [rad]

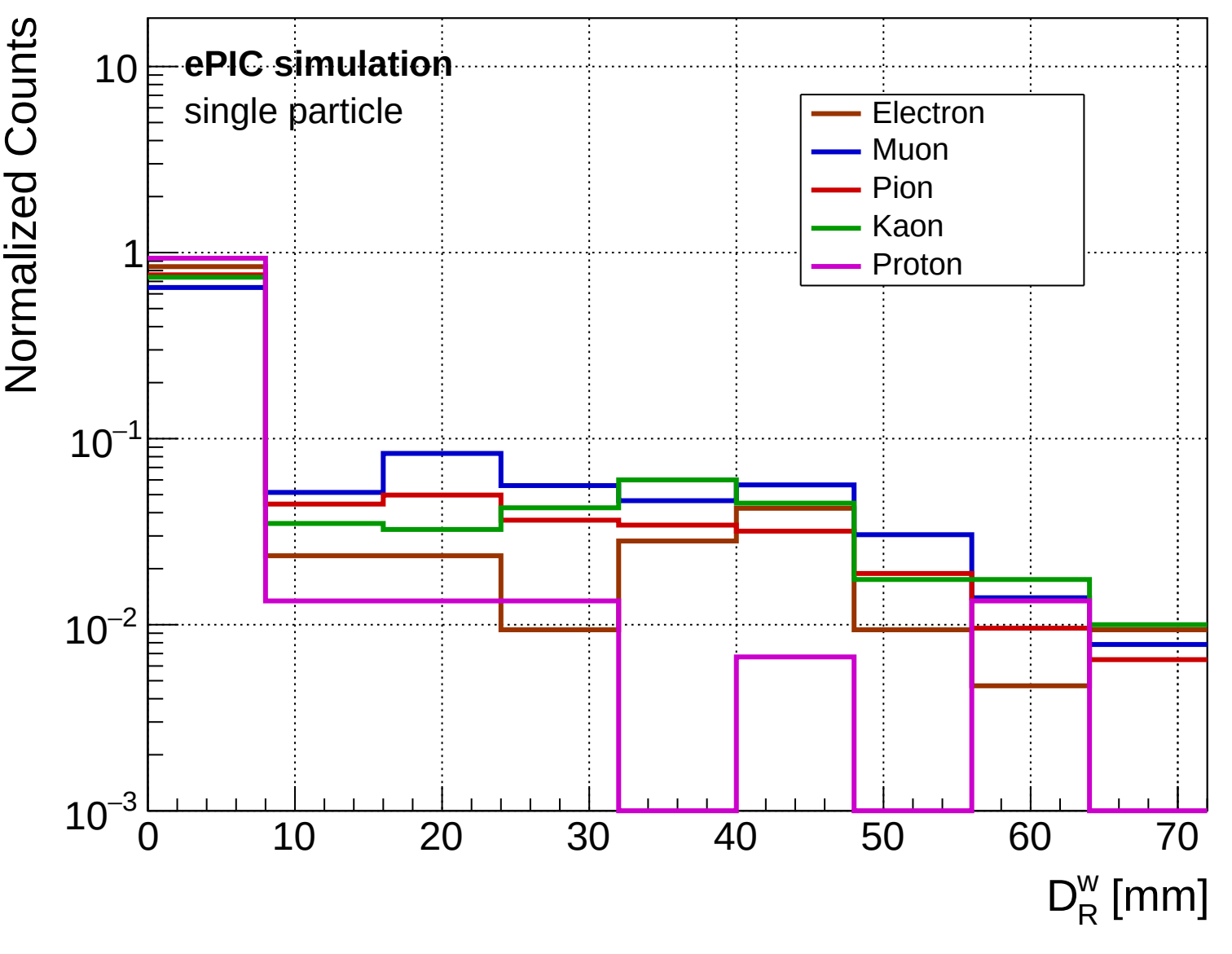

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.2
0.4
0.6
0.8
1
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
η_{Spread}

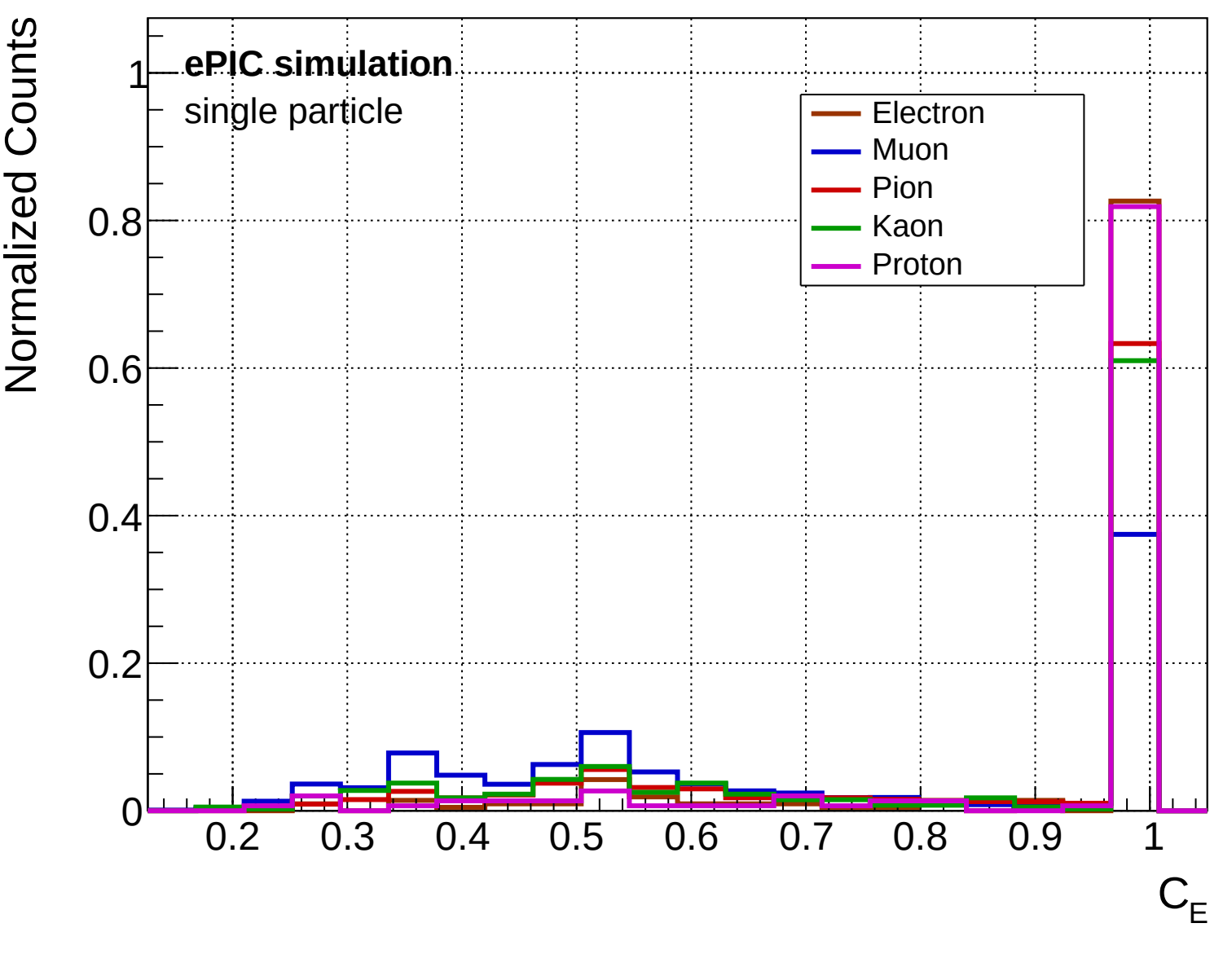

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
10
1
10^{-1}
10^{-2}
10^{-3}
0
50
100
150
200
250
R_{Spread} [mm]

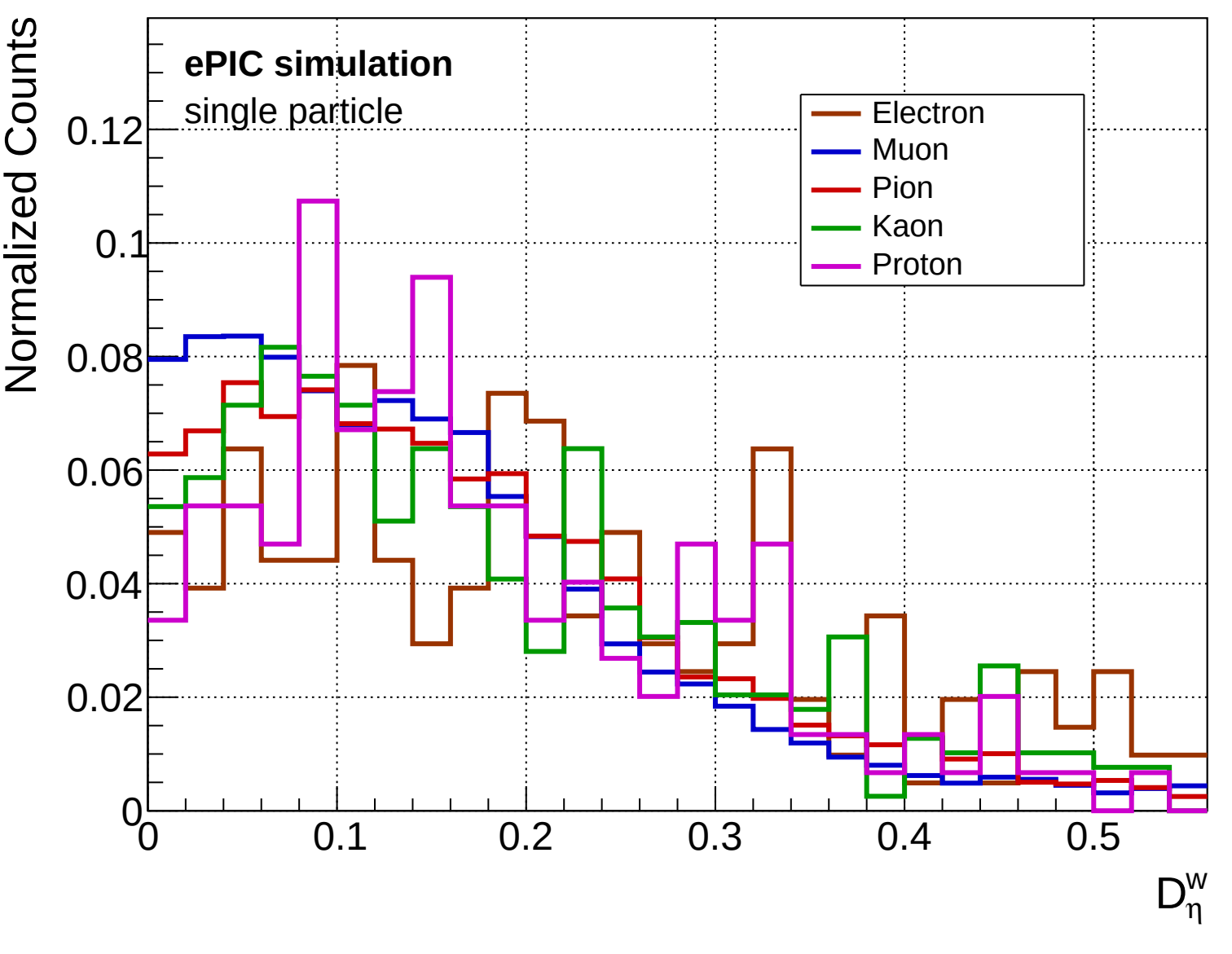

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.2
0.4
0.6
0.8
1
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
f_{max}

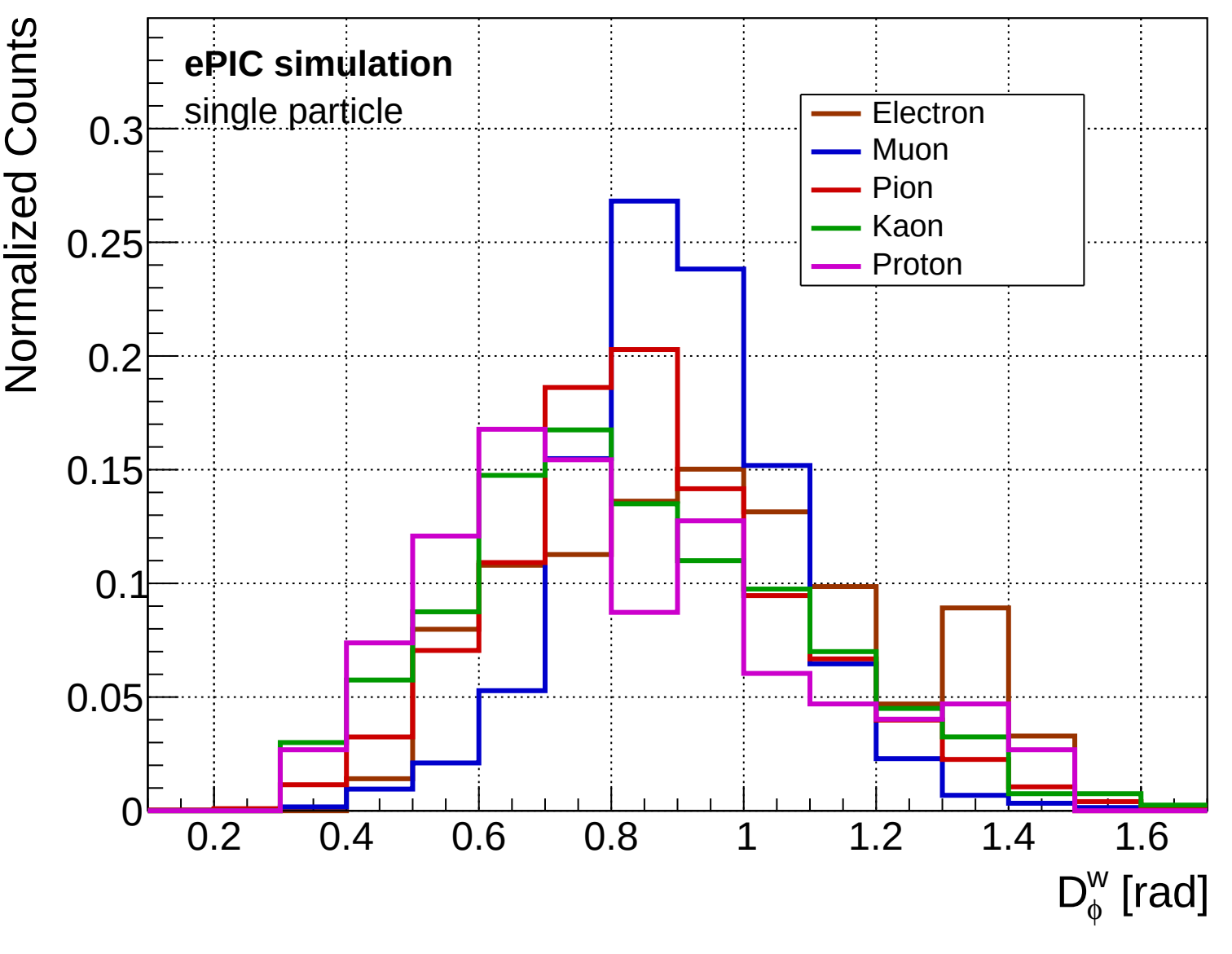

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
10
1
10^{-1}
10^{-2}
10^{-3}
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
σ_E [GeV]

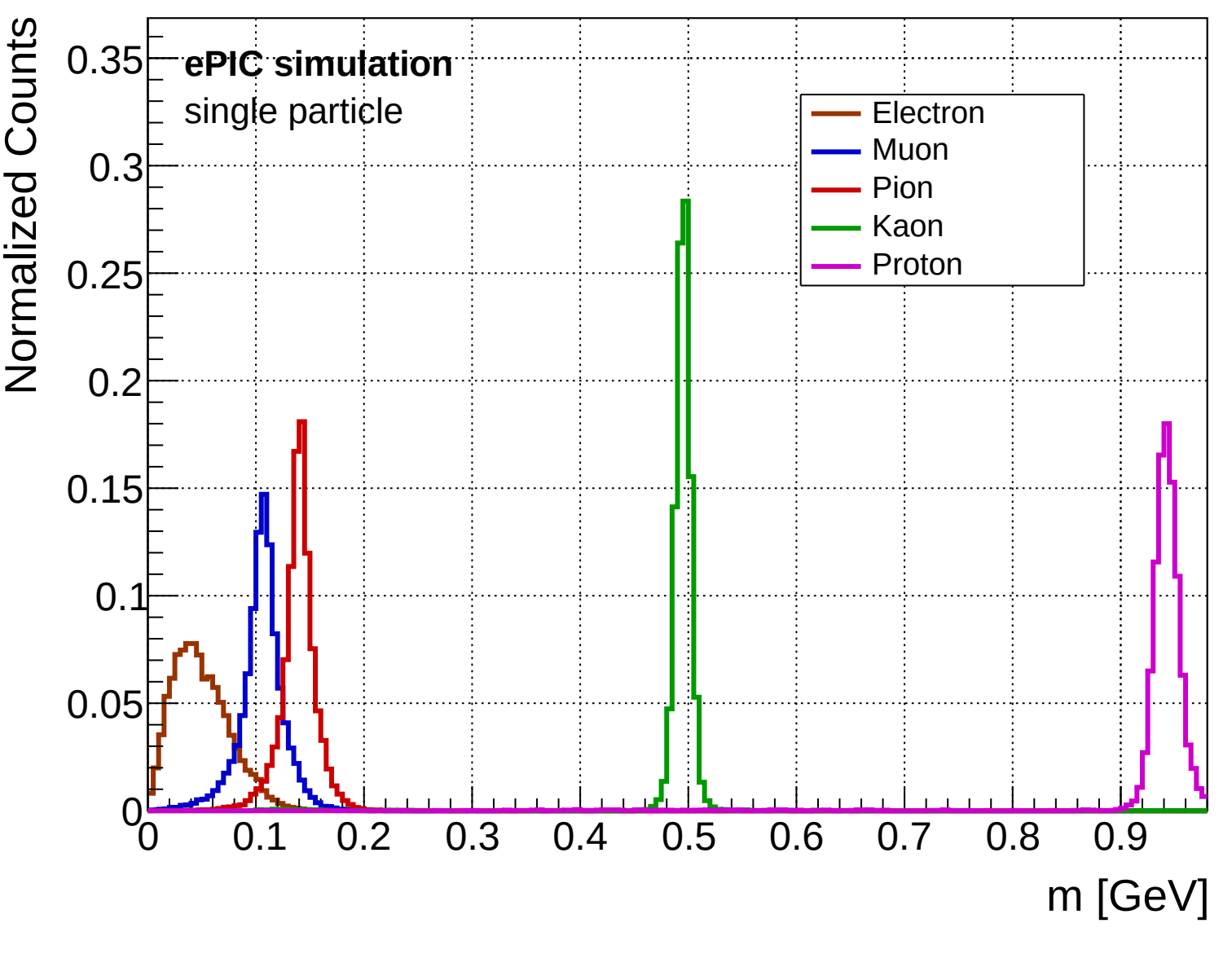

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
10
1
10^{-1}
10^{-2}
10^{-3}
0
20
40
60
80
100
D_R [mm]

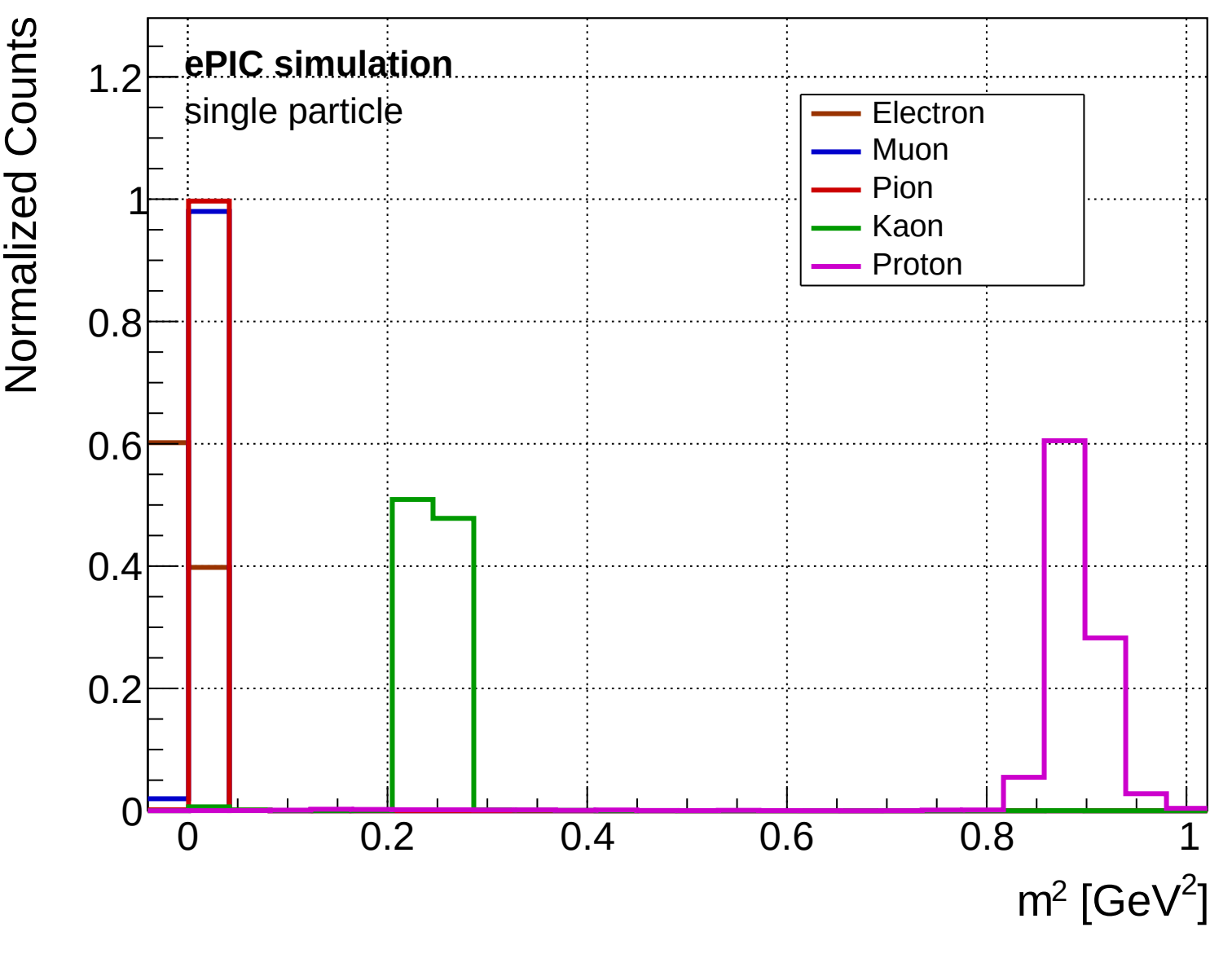

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
10
1
10^{-1}
10^{-2}
10^{-3}
0
10
20
30
40
50
60
70
D_R^w [mm]

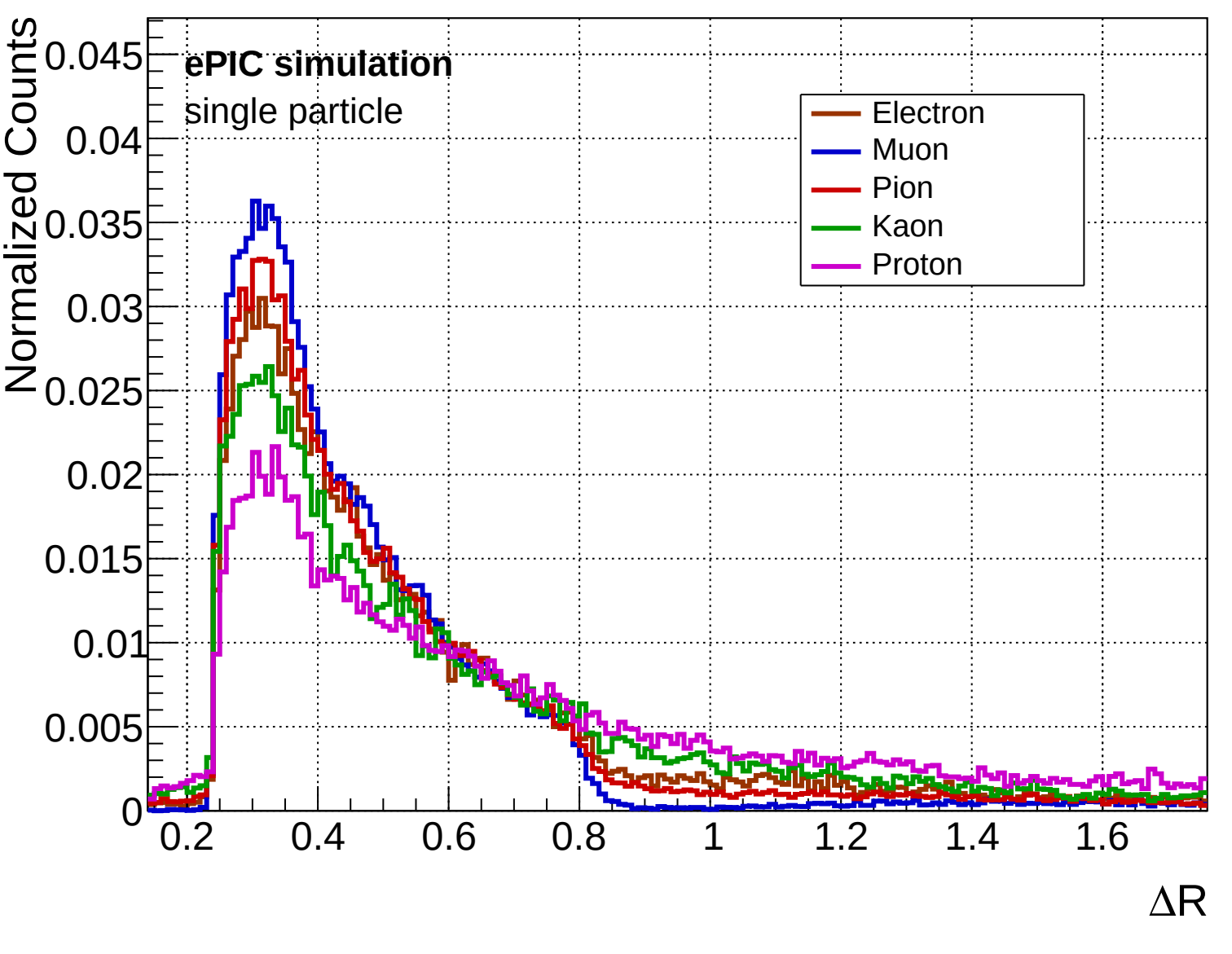

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.2
0.4
0.6
0.8
1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
C_E

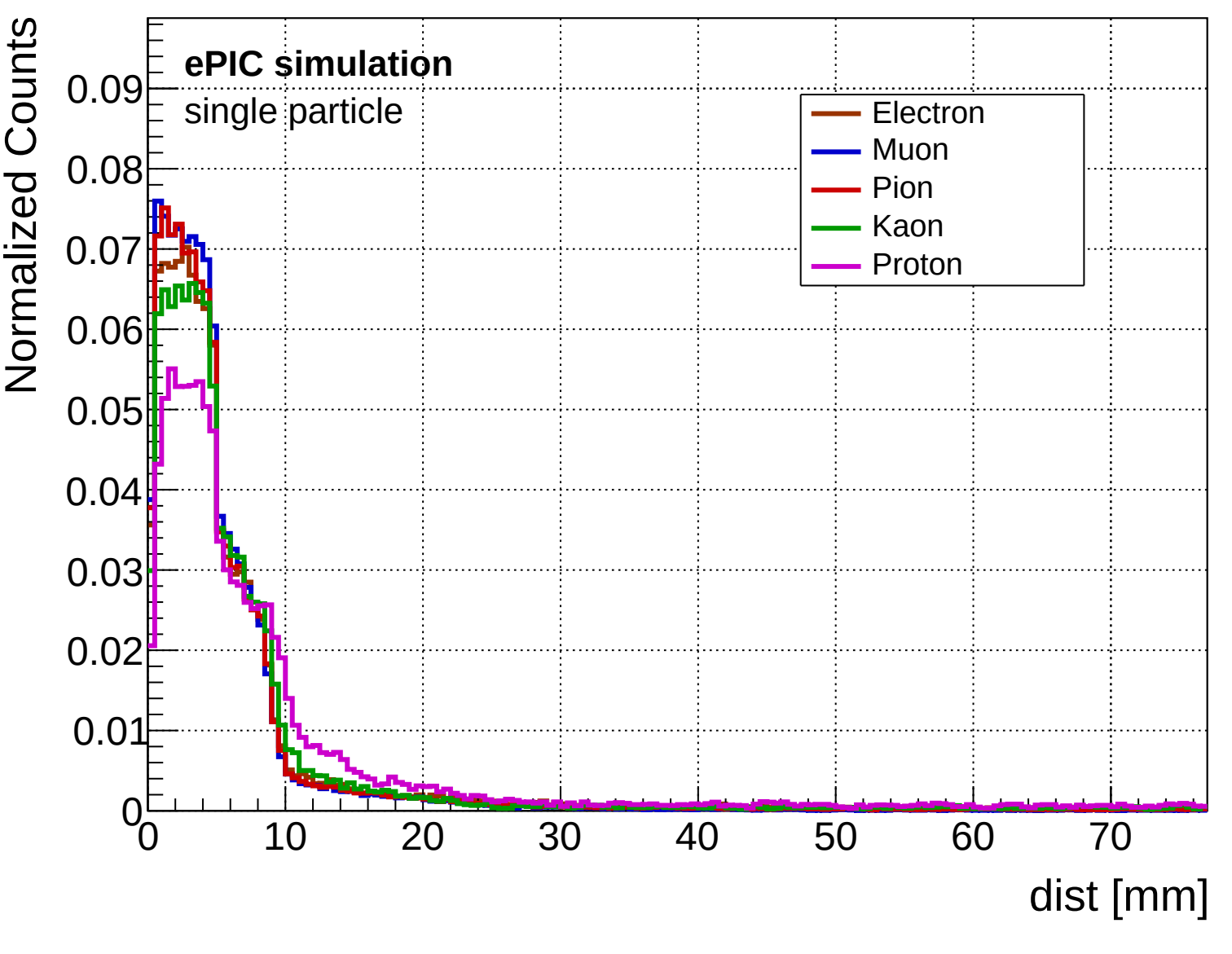

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.02
0.04
0.06
0.08
0.1
0.12
0
0.1
0.2
0.3
0.4
0.5
D_η^w

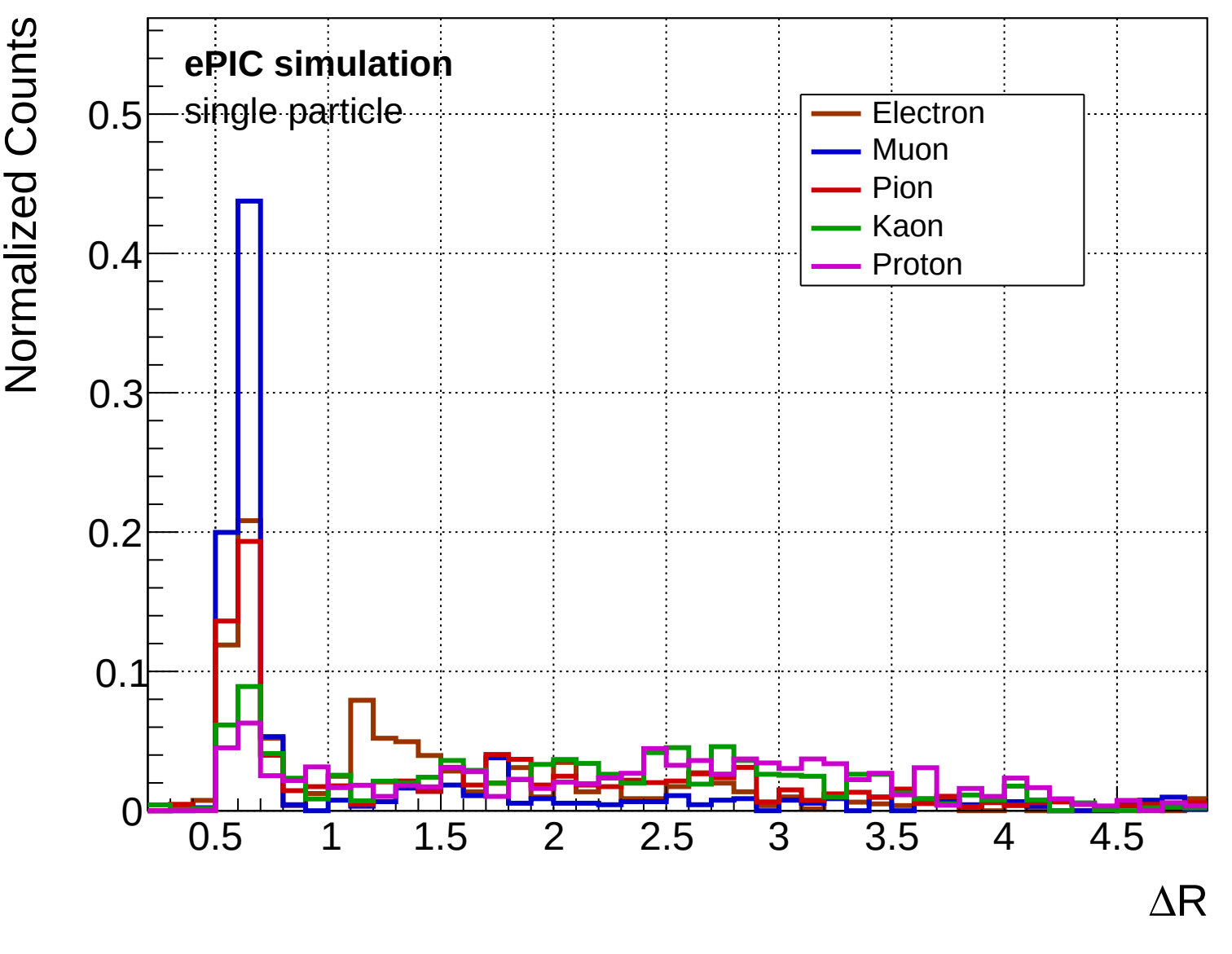

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.05
0.1
0.15
0.2
0.25
0.3
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
D_{ϕ}^{w} [rad]

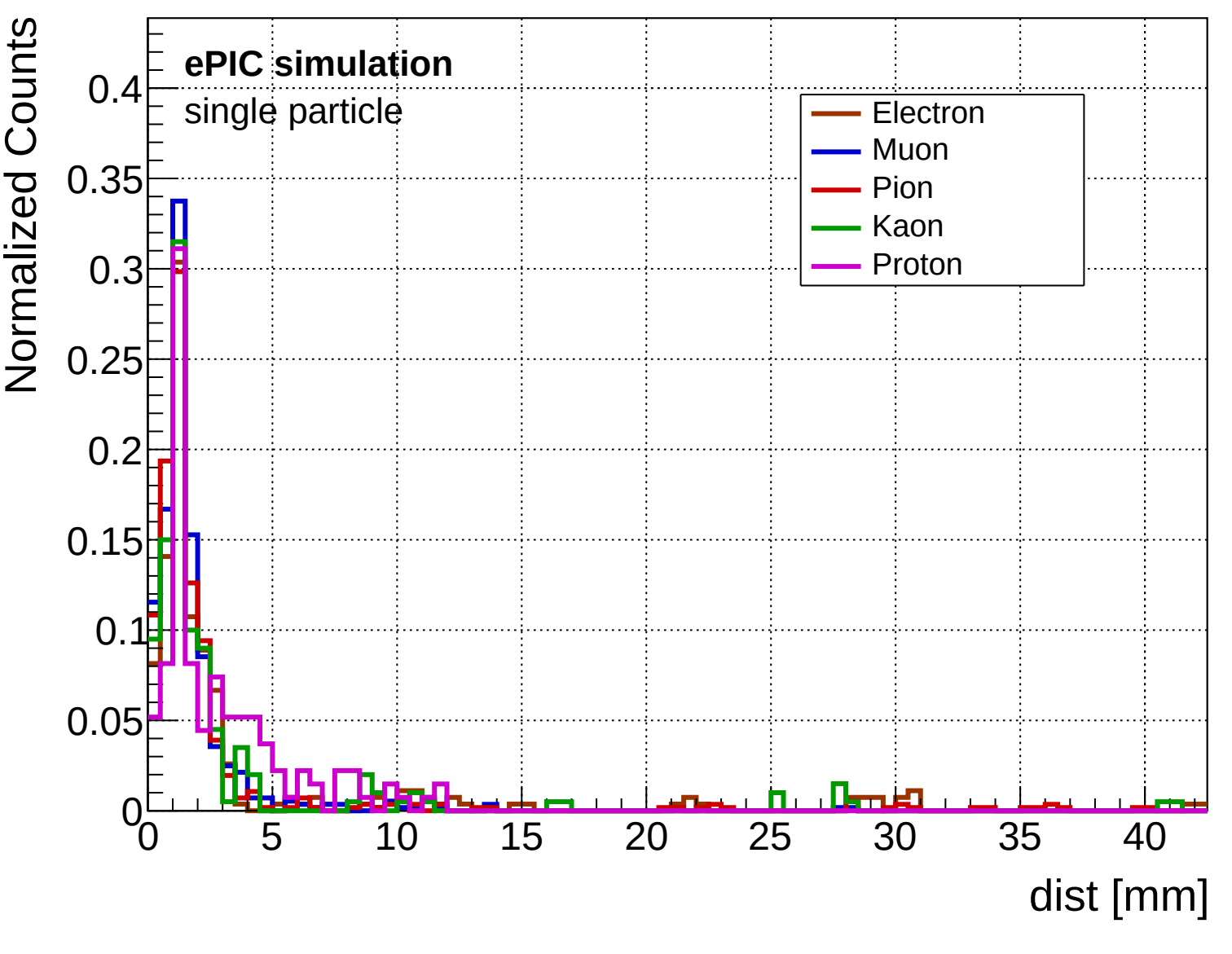

ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
m [GeV]


ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
1.2
1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1
m² [GeV²]


ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0.045
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
ΔR


ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0
10
20
30
40
50
60
70
dist [mm]


ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.1
0.2
0.3
0.4
0.5
0.5
1
1.5
2
2.5
3
3.5
4
4.5
ΔR


ePIC simulation
single particle
Electron
Muon
Pion
Kaon
Proton
Normalized Counts
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0
5
10
15
20
25
30
35
40
dist [mm]